MA Thesis Guide

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Welcome

Welcome to my **MA Thesis Guide for students of Communication Science** at the Vrije Universiteit Amsterdam. I designed this website to support you throughout your master's thesis project. Topics range from choosing a topic all the way to submitting your final version. It can be a somewhat daunting project and my hope is that this will help your sail smoothly through it.

This guide brings together information from several official documents (e.g., the thesis manual, writing guidelines, and departmental AI policy) and many more resources (e.g., github repositories that I helped creating) in hopefully one accessible, more searchable, and digestible format. Use it alongside my feedback during our supervision sessions as well as the general resources on Canvas (in case of doubt, check those carefully).

How to use this guide

This guide is organized into five parts:

1. **Getting Started**: Here you can find MA thesis course details, the thesis trajectory, key deadlines, how I envision the supervision.
2. **The Research Process**: How to move from research question to design, while taking ethics and data management into account, as well as and how to write the 2-page thesis plan and good research proposal (the first deliverables!)
3. **Writing Your Thesis**: A the core of this guide, you'll find section-by-section writing requirements and tips
4. **Academic Practices**: Here, you'll find general information about how to do a proper literature search, implement APA-conform citation and reference lists, engage with open science, AI tools, and using R for statistical analysis
5. **Submission & Graduation**: Finally, you'll also find information about submitting, grading, resits, and the final graduation

Use the sidebar to navigate and the search bar to find specific topics. You can also download the full guide as a PDF using the buttons in the top-right corner.

> New to the thesis process?
>
> Start with Getting Started for an overview of the course, then work through The Thesis Trajectory to understand what is expected at each stage. Pay close attention to the deadlines as well!

> Official documents
>
> Please note that this guide synthesizes and elaborates on the following official departmental documents (which you can find on Canvas):
>
> - *Manual for the MA thesis Communication Science 2025–26*

- *Guideline Masterthesis Communication Science*
- *Use of Generative AI Tools in Your Ba/Ma Thesis 2025–26*

In case of any discrepancy, the official documents on Canvas take precedence!

About this guide

This guide was created by **Philipp Masur** (Associate Professor in the Department of Communication Science at the VU Amsterdam). I will update it frequently to reflect current departmental policy and best practices as well as include more resources that I deem helpful.

If you notice an error or want to suggest an improvement, please "report an issue" via the link in the right margin of the page.

Part I.

Getting Started

1. Getting Started

Beginning with a MA thesis can be a bit intimidating. It is likely the biggest (academic) project that you'll ever go through and unfortunately, you'll have to do it more or less by yourself. But I am here to help and guide you. As a first step, I suggest to familiarize yourself with the formalities below.

1.1. Course information

The MA Thesis in Communication Science is a substantial independent research project that represents the culmination of your master's degree. Here are the key course facts:

Course code	S_MTcw
Credits	18 EC (504 hours)
Course level	600
Periods	4–6 (February track) or 1–2 (September track)
Language	Dutch or English
Faculty	Faculty of Social Sciences
Teaching format	Individual and small-group meetings with me, plus centralized support and student peer review
Assessment	Evaluated by me (first reviewer) and a second reviewer using the assessment form (see Canvas)

1.2. Entry requirements

Before you can actually begin with the full MA thesis trajectory, you must meet two requirements:

1. A minimum of **12 EC** in total from preceding MA courses, including successful completion of **Research Methods for Communication Science**
2. An **approved Research Proposal** after the first phase (see The Thesis Trajectory)

1.3. Start dates: February or September track

The thesis course formally starts in **early February** (Period 4). However, it is also possible to start after the summer, in **early September** (Period 1).

Why choose the September track?

- You are doing an internship or other study activity that does not fit alongside the thesis
- You did not pass all courses from Period 1 or 2

If you want to start in September, discuss this with me during the thesis market in January. Note that only a few supervisors offer supervision in the September track. You must still participate in the January supervisor assignment procedure.

1.4. Registration

Preparation begins in **December** with an information lecture about procedures (check your mails!), deadlines, and available supervisor topics. **Register for the course before December** to receive information via Canvas.

> **!** Don't miss the information lecture
>
> The information lecture (usually early December) is where you receive the full list of supervisor topics and procedural information. Missing it can mean missing important deadlines. This is also your chance to learn about research topics and interest by different supervisors!

1.5. Choosing a topic

In **January**, a **thesis market** is organized where you can walk around and talk informally with different supervisors about their research themes. Please come and visit me as I can explain my research interests and topics in much more detail there.

After the market, you submit an **online questionnaire** with your topic preferences (ranked 1–8). If you would like to work with me, you need to put my topics high in your ranks and at best also indicate in the comment section that you would like to work with me.

Supervisor assignments are generally finalized by the **end of January**.

1.5.1. Can I propose my own topic?

In general, I would like you to work on the topics that I proposed during the information lecture and the MA thesis market. Yet, I am open to good research ideas. A good own proposal should contain at minimum:

- A research question
- A brief theoretical foundation
- The intended type of research (survey, experiment, content analysis, etc.)
- Indicate why you belief it fits within my research interests

1.5.2. Can I write my thesis in English?

Yes. In fact, I prefer if you would write your thesis in English, but I also allow for you to write in Dutch.

1.6. Getting organized

A thesis spans several months and involves dozens of papers, multiple drafts, and a growing analysis folder. Students who set up a simple system at the start save themselves significant stress later.

Reference manager — use Zotero from day one

Zotero (free) lets you save references directly from your browser, organize them into collections, and generate APA citations automatically in Word or R Markdown. Set it up before you start reading. See Literature Search for more detail.

Take notes as you read

When you read a paper, immediately write down the key argument, the main findings, and how it connects to your own study. Do not rely on being able to recall this later. A simple document per topic or a note in Zotero is enough — but do it while the paper is open in front of you.

Break the thesis into SMART goals

"Write the theory chapter" is paralyzing. "Draft the first paragraph of the social norms section by Thursday" is not. Break every phase of the thesis into small, specific, time-bound tasks. This is the single most effective way to maintain momentum on a long project.

Back up your work

Save your files to OneDrive or another cloud service automatically and continuously. Also keep your analysis syntax and data in the shared SURFdrive folder from the start (see Ethics & Data Management) — not only at the end.

2. The Thesis Trajectory

The MA thesis project consists of three main phases: **preparation (P4)**, **research and writing (P5)**, and **finalizing and submission (P6)**. In this chapter, I explain what is expected at each stage.

2.1. Phase 1 (P4): Preparation

Period 4 runs from **February to late March**. During this period, the thesis course runs in parallel with your two elective courses. You should plan to spend approximately **4 hours per week** on your thesis in this phase. It is crucial to develop first ideas and narrow down your focus. You will meet several times with me to discuss and brainstorm.

Your main tasks in P4:

- Read relevant literature in your chosen area
- Delineate your individual research focus
- Develop a draft research question
- Meet with me (and your thesis group) to get feedback on your ideas

Start writing immediately

As soon as you start reading, do it systematically: collect articles, write down your ideas and summaries. Keep an overview of all sources (e.g., using citation managers such as Zotero) and note the relevant aspects of each paper. This prevents you from getting lost in the literature and speeds up writing later. A good overview of the literature is key to not only an interesting and focused research question, but also for writing a large part of your thesis later on.

2.1.1. The MA Thesis Plan (end of P4)

By the **end of Period 4** (see Timeline & Deadlines), you submit a so-called **Thesis Plan** (sometimes also called **Exposé**) to me via email (see also Writing the Thesis Plan). This is a **minimum two-page document** that outlines your proposed project. The further you have developed it, the better. It should include at minimum:

1. A first description of the relevant literature and core theory, and a research "gap" that justifies your focus (theoretical and societal relevance)
2. Your over-arching research question, i.e., what you would like to answer with your study
3. Related hypotheses (if the study is confirmatory) or sub-questions (if it is exploratory)
4. A description of your intended research design (e.g., *"a survey exploring the relationship between X and Y"*, or *"a 2×2 between-subjects experiment with X as IV and Y as DV"*)

Think of the Thesis Plan as a blueprint for the next step, your Research Proposal: (1) and (2) become your Introduction, (3) goes into the Theory section, and (4) goes into the Method section.

2.2. Phase 2 (P5): Research and Writing

Period 5 begins in April. From now on, you should work **full time** on your thesis.

2.2.1. The Research Proposal (early P5)

In the **first two weeks of P5**, you should intensively develop your **Research Proposal**, building on the Thesis Plan and incorporating my feedback. The Research Proposal must contain:

1. **Introduction** — including a clearly formulated over-arching Research Question
2. **Theory** — theoretical grounds and argumentation for the relationships or effects you expect
3. **Final hypotheses** (or sub-questions for exploratory or qualitative research)
4. A **conceptual model** (for quantitative research)
5. **Method** — research design, (in)dependent and control variables, ideas on procedure and operationalization (a first draft of the questionnaire, codebook, etc. can go in the appendix)
6. **Analysis plan** — for each hypothesis/RQ, specify exactly which statistical test on which variables you will use

! The Research Proposal is a go/no-go moment in your thesis trajectory

I evaluate the proposal as *sufficient* or *insufficient*. An **approved** Research Proposal is required to continue the thesis trajectory. If it is rejected, you have up to two resits (within 10 working days each). Failing both resits means you must stop and restart in the following academic year.

Upload your Research Proposal on Canvas and email it to me before the deadline. Give peer feedback to fellow students first, and incorporate their feedback before submitting.

2.2.2. Complete Methodology

As soon as possible after the Research Proposal is approved, submit your **complete methodology**. This is the final, detailed operationalization of your method, which may include the exact content of your questionnaire, experimental materials, codebook, Qualtrics setup, etc. A lot this information will be included in the appendix of your final thesis.

2.2.3. Ethical Review

Before you start collecting data, you must complete the **ethical self-check** via the Research Ethics Review Committee (RERC). See Ethics & Data Management for details. You will go through this with me together.

2.2.4. Data Collection and Analysis

Once your Research Proposal and Complete Methodology are approved and the ethical self-check is done:

- You will start **data collection** (often in consultation with me, depending on your project)
- Write the **Method chapter** (in past tense)
- Conduct analyses and write the **Results chapter**

> Prepare analyses before data comes in
>
> You can write your analysis syntax (R or SPSS) and structure your results section before the final data is collected. This saves a lot of time.

2.3. Phase 3 (P6): Finalizing and Submission

Period 6 is the final stretch. After receiving feedback on your Method and Results chapters, you:

- Write the **Conclusion and Discussion** chapter
- Polish Introduction, Theory, Method, and Results
- Process all remaining feedback
- Submit the **complete concept version** to me for final feedback (see deadlines)
  - This round focuses on: (1) overall coherence, (2) the discussion/conclusion section, and (3) any specific questions you have
- After processing all feedback, submit the **final version**

> Quality matters — for you and for me
>
> Submit only high-quality draft versions, meaning: versions you have already peer-reviewed with fellow students and improved. Feedback can only be useful and focused if the text is already in good shape.

2.4. Peer feedback

Throughout the trajectory, you are expected to give and receive **peer feedback** with fellow students in your thesis group. Peer feedback is required before submitting the following to me:

- The Research Proposal
- The Method and Results chapters
- The complete concept version

Use the official **Guideline Masterthesis Communication Science** (Canvas) as your peer feedback checklist.

3. Timeline & Key Deadlines

On this page, you'll find a rough overview of the timeline and important deadlines. Some of them are fixed and you cannot miss them. Otherwise are more of a rough estimation of when certain milestones should happen.

3.1. Monthly overview

The table below shows the typical tasks and tips for each month in the **February track** (P4–P6). The same structure applies to the September track, shifted by approximately seven months.

Month	Tasks	Tips
December	Thesis information lecture	Think about what you want to study. A preliminary list of supervisor topics is published before Christmas.
January	Thesis market — meet supervisors, discuss topics, submit preferences questionnaire	Come prepared and with an open mind. Talk to a range of supervisors to find the right fit.
February (P4 starts)	Read relevant literature; delineate your individual focus; develop a draft research question	Start writing down your ideas immediately. Use Zotero to keep track of sources.
March	Determine your research gap; formulate your research question; think about your method. Submit **MA Thesis Plan** (2-pager) to me by end of P4.	Everything you do in P4 is preparation for a flying start in P5.
April (P5 starts)	First two weeks: intensively develop your **Research Proposal** (intro, theory, method, analysis plan). Process peer feedback before submitting.	Use the knowledge clips and support resources on Canvas for writing and hypothesis formulation.

Month	Tasks	Tips
April	Work out your **Complete Methodology**. Once Research Proposal + analysis plan + methodology are approved: do the ethical self-check, and optionally submit a pre-registration. Then start data collection.	
Late April	Start data collection. Write method chapter; conduct and report analyses in results chapter.	Keep a clean SPSS/R syntax from the start. You can prepare your analysis syntax before data is in.
Early May	Process peer feedback on method and results chapters before submitting to me.	Relate results to hypotheses and theory; seek additional literature if needed.
May	Write and polish theory, method, results; write discussion and conclusion. Process peer feedback on complete concept version before submitting.	Writing a good thesis always requires many revisions.
Early June	Submit complete concept version to me. Process feedback and refine.	Agree with me in advance when to submit — this is a busy time for everyone.
End of June	Submit **final version**	See deadline table below.

3.2. Deadline tables (2025–26)

3.2.1. February track

Event	Date
Information lecture	Tuesday, 2 December 2025
Thesis market	Tuesday, 13 January 2026
MA Thesis Plan deadline	Tuesday, 31 March 2026
Research Proposal deadline	Tuesday, 14 April 2026
1st submission deadline	**Friday, 26 June 2026, 16:00**
Resit deadline	Friday, 24 July 2026, 17:00
2nd resit (if allowed)	Friday, 20 November 2026

3.2.2. September track

Event	Date
Supervisor assignment	January 2026
Start of trajectory	Beginning of September 2026

Event	Date
MA Thesis Plan deadline	Monday, 19 October 2026
Research Proposal deadline	Monday, 2 November 2026
1st submission deadline	**Friday, 15 January 2027, 17:00**
Resit deadline	Friday, 19 February 2027, 17:00
2nd resit (if allowed)	Friday, 18 June 2027

! Missing the first deadline = missing an exam opportunity

Not submitting at the first opportunity is treated the same as not showing up for an exam. You are then entitled to one resit. If you fail the resit as well, you have exhausted your attempts for the current academic year.

3.3. My availability

I am available for supervision meetings during the regular thesis period, **up to and including the first submission deadline**. During July and August**, I am generally not available for feedback.

Plan accordingly. Do not rely on my feedback during the summer months.

4. Supervision & Peer Feedback

Supervision is generally very project-specific. My general approach is one of **scaffolded independence**: early in the project, when the research question and design are still being developed, I provide more structured guidance and frequent feedback. As the project matures and you gain confidence, I step back and you take the lead — I become a sounding board rather than a co-pilot. The goal is that by the time you submit, you are capable of designing and executing rigorous research independently.

In practice: if you are highly independent and motivated, I will guide you lightly and otherwise leave you to work. If you feel lost or stuck, I will support you more. Below you find how I usually structure supervision across the trajectory.

4.1. What you can expect from me as your supervisor

I will guide you in:

- Formulating an interesting and well-grounded research question and hypotheses
- Writing a coherent, well-structured, and theoretically sound thesis
- Designing a proper methodology and analysis plan
- Giving focused feedback on each chapter (**once per chapter**)

As a rule of thumb, I will provide feedback on:

1. Your **introduction** (as part of the Thesis Plan feedback)
2. Your **theory chapter and research design** (as part of the Research Proposal feedback)
3. Your **complete methodology** (right before data collection=)
4. Your **method and results chapters**
5. The **complete concept version** (focus: coherence and discussion/conclusion)

4.2. What I expect from you

A master's thesis student is expected to demonstrate the capacity to **independently** set up, conduct, and report an empirical research project. I am a guide and supervisor — not a co-author.

This means:

- Show **initiative**: contact me with a clear agenda, not just general questions
- Submit **high-quality drafts**: only submit versions that you (and your peers) cannot improve further on your own
- **Do not ask me* about basic matters such as how to conduct a literature search, spell-checking, APA formatting, how to run R/SPSS tests, or things you have already been taught in your coursework (many answers can also be found on this website!)

For technical analysis questions (how to run tests in R or SPSS, how to report results), use:

- The statistics helpdesk
- The *R for Social Scientists* Canvas page
- The departmental knowledge clips on Canvas
- The Guideline Masterthesis Communication Science document

> ⚠ Feedback is only useful if you've done your part
>
> Feedback is pointless on text you know has flaws. Submit drafts only when you have already done everything you can to improve them, including peer feedback.

4.2.1. What level of thinking is expected?

A useful way to think about intellectual expectations is the **SOLO taxonomy** (Biggs & Collis, 1982), which describes five levels of understanding:

Level	Description	Typical sign
Pre-structural	No real understanding	Misses the point; irrelevant content
Uni-structural	Can identify or recall single facts	Defines concepts correctly but does nothing with them
Multi-structural	Can list or describe multiple elements	Summarizes several studies without connecting them
Relational	Can link, compare, and integrate	Builds a coherent theoretical argument; connects findings to hypotheses
Extended abstract	Can generalize and theorize	Situates findings in broader debates; proposes new theoretical directions

A passing MA thesis demonstrates **relational** thinking throughout — you connect theory to empirical evidence, link findings to hypotheses, and build a coherent argument. An excellent thesis reaches **extended abstract** thinking in the discussion — you generalize beyond your data, consider alternative explanations, and make a genuine theoretical contribution. Simply summarizing what others have found (uni- or multi-structural) is not sufficient at this level.

4.3. Thesis groups and peer feedback

At times, I organize supervision in **thesis groups** of 2–4 students working on related topics or even collecting data together. If that is the case, you will meet together especially in P4. As projects become more individualized, groups may split into smaller subgroups or even individual meetings. Across the full trajectory, expect approximately **6–8 meetings** with me.

4.3.1. Peer feedback process

Before submitting any draft to me, you must:

1. Give and receive peer feedback with fellow students
2. Process the peer feedback and improve your draft
3. **Then** submit the improved version to me

Peer feedback is required for:

- The **Research Proposal** (including introduction and theory)
- The **Method and Results chapters**
- The **complete concept version** of the thesis

Use the **Guideline Masterthesis Communication Science** (Canvas) as your peer feedback checklist. As a peer reviewer, reflect on each section in a few sentences and suggest concrete improvements.

4.4. Formal regulations

The formal regulations for the MA thesis are described in the *Regulation Ma thesis* document, available on VUnet. In practice, we can usually find a good working relationship without needing to consult the formal rules. However, in cases of disagreement, the regulations serve as the official reference point.

4.5. Getting help for personal circumstances

If personal circumstances are affecting your progress:

- Contact the **study advisor** as early as possible: vu.nl/en/student/contact-student-guidance-and-support/academic-advisors-fss
- For delays due to illness, psychological issues, or other circumstances, you may be eligible for financial support from the **Profileringsfonds**: vu.nl/en/student/finances/profile-fund

Report delays early

The sooner you contact your study advisor about a delay, the more options are available to you. Waiting too long limits what support can be offered.

Part II.

The Research Process

5. From Topic to Research Question

5.1. Starting with a topic

A thesis begins with a **topic of interest**, a broad area within communication science that you want to explore. I offer research themes within my expertise. Working within my themes ensures your project is relevant and that you benefit fully from my expertise. My key research areas include:

1. Privacy and Self-Disclosure Dynamics in various Online Environments
2. Social Norms and Social Influence on Social Media
3. Digital Literacy (incl. e.g., Privacy Literacy, Algorithm Literacy, and AI literacy)

Beyond those core topics, I am also willing to supervise topics related to the following areas:

4. Consequences of using Generative AI for Individuals and Society
5. Social Media Use Effects on Individual Well-Being

To gain a better understanding of my research, have a look at my publications.

If you propose your own topic, make sure it is:

- **Interesting and feasible** within the timeframe
- **Anchored in existing literature** (you must be able to find a research gap)
- **Researchable** with data you can realistically collect
- Something I can guide — it should fall within my research interests

5.2. Identifying the research gap

A good research question emerges from a **gap in the literature**. After reading the relevant literature in your area, ask yourself:

- What do we know already?
- What is still unclear, inconsistent, or unexplored?
- Why does this gap matter — theoretically and/or for society?

The gap is your justification for doing the study. It should be clearly articulated in your Introduction and elaborated in your Theory chapter.

5.3. Formulating the research question

A good research question is:

- **Specific** — it names the constructs/variables you are interested in
- **Researchable** — it can be answered with empirical data you can collect
- **Theoretically grounded** — it follows from a gap in existing knowledge
- **Focused** — one central question, not multiple questions combined

Common mistake: too broad

"What is the role of social media in society?" is too broad. A better version: "To what extent does exposure to ideologically congruent social media content predict political polarization among Dutch adults?"

In **quantitative, confirmatory research**, the main RQ is typically followed by **hypotheses** that predict specific relationships. These are derived from existing theory.

In **qualitative research** (e.g., grounded theory, thematic analysis) or **exploratory, quantitative research**, the main RQ is usually followed by **sub-questions** that guide your data collection and analysis.

5.4. Hypotheses

A hypothesis proposes a specific, testable relationship between variables. Good hypotheses:

- Propose **one effect** or **relationship** (not multiple effects/relations in one statement)
- Follow **logically** from the theoretical argument
- Are supported by theory and/or **3–5 previous studies** pointing in the same direction
- Use **consistent terminology** with the theory section, conceptual model, and method section that further aligns with the existing terminology in the field

Example: *"H1: Higher exposure to health misinformation on social media will be positively associated with vaccine hesitancy."*

More examples from typical research areas in Communication Science:

Research area	Example hypothesis
Privacy & self-disclosure	*"H1: Higher privacy literacy will be negatively associated with self-disclosure depth on social media."*
Social norms	*"H2: Exposure to descriptive norms showing high peer disclosure will increase participants' own disclosure intentions."*
Moderation	*"H3: The positive effect of peer disclosure exposure on disclosure intentions will be stronger for users with lower privacy concerns."*
Mediation	*"H4: Privacy concerns will mediate the negative association between privacy literacy and self-disclosure."*

Research area	Example hypothesis
Digital literacy	*“H5: Higher algorithm literacy will be negatively associated with susceptibility to algorithmically curated misinformation.”*
GenAI use	*“H6: More frequent use of generative AI tools for work tasks will be positively associated with feelings of work-related dependency.”*

! Common problem: Non-falsifiable hypotheses

Hypotheses must be falsifiable, i.e., it should be possible that the data does not support them. For example, a hypothesis like this *“Exposure to health misinformation has an effect on vaccine hesitancy”* is almost not falsifiable as it doesn’t specify a direction and thus any found effect would support it.

! Common problem: Hypotheses that combine multiple effects

The following hypothesis looks specific but is actually two separate predictions in one statement:
“Privacy literacy is negatively associated with self-disclosure, and this effect is stronger for women and older users.”
This should be split into three hypotheses: one main effect (H1) and two moderation effects (H2, H3). Main effects and moderation/mediation effects must always be stated in separate hypotheses.

5.5. Sub-questions (qualitative research)

In qualitative projects, the main RQ is broken down into **sub-questions** that:

- Follow logically from the main RQ
- Address specific aspects or steps of your research
- Are likewise grounded in 3–5 previous studies
- Are formulated distinctively (each addresses a clearly different focus)

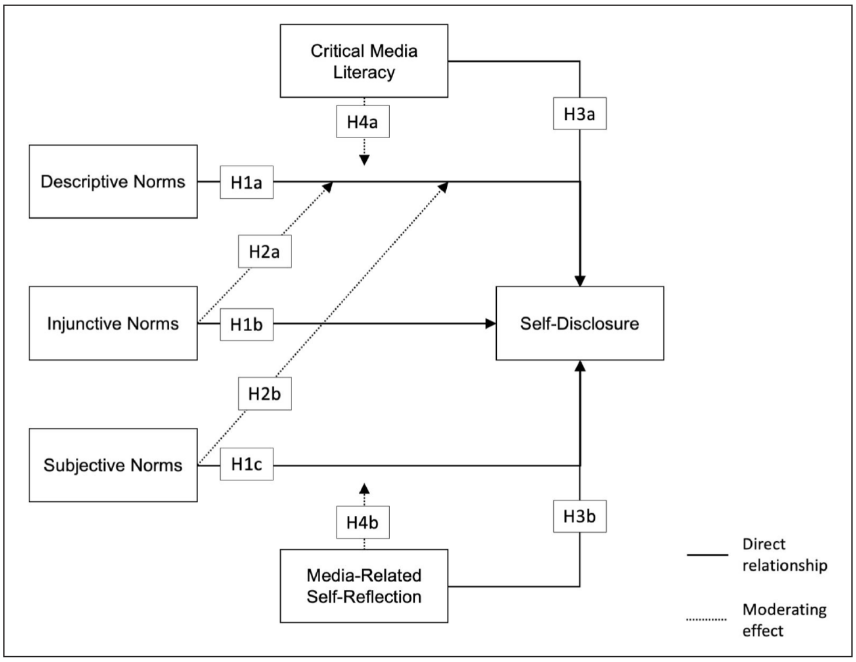
Each sub-question ends with a **single question mark** — a common mistake is combining multiple questions before one question mark.

5.6. The conceptual model

For quantitative research, your Theory chapter ends with a **conceptual model** — a visual diagram that maps out the constructs and their hypothesized relationships. Every variable mentioned in your hypotheses must appear in the model. Every variable in the model must appear in your hypotheses and method section.

Use consistent terminology throughout: the same construct should have the same name in the introduction, theory, conceptual model, method, and results.

Example conceptual model:



Source: Masur, P. K., Bazarova, N. N., & DiFranzo, D. (2023). The impact of what others do, approve of, and expect you to do: An in-depth analysis of social norms and self-disclosure on social media. Social Media + Society, *9*(1), 1–14. https://doi.org/10.1177/20563051231156401*

6. Research Design

Choosing the right research design is one of the most important decisions you make in your thesis. Your design must fit your research question. This chapter outlines the main options available to you.

6.1. Quantitative vs. qualitative research

	Quantitative (Confirmatory)	Qualitative
Goal	Test hypotheses about relationships between variables	Explore meanings, experiences, patterns
Data	Numerical (survey responses, coded content)	Text, images, audio (interviews, documents)
Analysis	Statistical tests (regression, ANOVA, etc.)	Thematic analysis, grounded theory, discourse analysis
Output	Generalizable findings across populations	Rich, contextualized insights about individuals
Thesis structure	Hypotheses + conceptual model	Sub-questions

Both approaches are valid in Communication Science. The choice depends on your research question and the state of knowledge in your area. I generally supervise quantitative research, but am open to qualitative or mixed-method approaches as well.

6.2. Survey research

A survey measures variables via a questionnaire administered to a sample of participants. It is the most common design in Communication Science master's theses.

When to use it: When you want to assess the prevalence of variables and/or examine associations and relationships between them in a larger sample.

Key considerations:

- Sample size and recruitment (you need to conduct a power analysis beforehand, see e.g., this blogpost)
- Validated scales vs. self-constructed items
- Platform (Qualtrics is the standard at VU)
- Online convenience sampling vs. probability sampling

> **Self-report validity**
>
> Self-reported media use correlates only r .38 with device-logged behavior — below the convergent validity threshold (Parry et al., 2021). If *actual* use (not *perceived* use) is central to your research question, consider whether self-report is a sufficient proxy. Always label self-reported use explicitly as such and qualify your inferences accordingly.

> **AI bots in online surveys**
>
> Recent research estimates that 30–90% of responses in crowdsourcing platforms (e.g., Prolific, MTurk) may be inauthentic, with AI-generated responses reaching up to 45% in some studies (Panizza et al., 2026). Even 3–7% polluted data can distort results enough to invalidate conclusions. When recruiting online, enable paradata tracking (response time, keystroke patterns) in Qualtrics, include attention checks, and discuss your data quality approach with me before launching your survey.

6.3. Experimental research

An experiment manipulates one or more independent variables and measures effects on dependent variables, with random assignment of participants to conditions.

When to use it: When you want to establish causal effects.

Key considerations:

- Sample size and recruitment (you likewise need to conduct a power analysis beforehand, see e.g., this blogpost)
- Online convenience sampling vs. probability sampling
- Stimulus design and setting
- Use of deception (should participants know about the purpose of the study?)

Design notation: Describe your design clearly, e.g.:

> *"A 2 (color of logo: red vs. blue) × 2 (medium: TV vs. newspaper) between-subjects experiment with purchase intention as the dependent variable."*

Always specify:

- Whether it is between-subjects, within-subjects, or mixed
- Whether randomization was used (it should be)
- Whether you conducted a **manipulation check**

6.4. Content analysis

Content analysis systematically codes and analyzes a corpus of media content (text, images, video, social media posts).

Types:

- **Manual content analysis**: human coders apply a codebook; requires intercoder reliability testing

- **Automated content analysis**: computational methods (e.g., dictionary-based, supervised machine learning, LLMs) via R, AmCat, or Coosto

Key considerations:

- How was the corpus assembled? (sampling strategy)
- What is the unit of analysis?
- How were variables operationalized and coded?
- What are the reliability and validity of your coding?

Example corpus description:

> *"We collected all articles published by the five most-read Dutch online news outlets between January 1 and March 31, 2024, using the Coosto API. After removing duplicates and non-news items (e.g., job listings), the final corpus consisted of 12,847 articles. The unit of analysis was the individual article."*

6.5. Qualitative research

Qualitative designs explore social phenomena in depth without the goal of statistical generalization.

Common approaches in Communication Science:

- **Interviews** (semi-structured, in-depth)
- **Focus groups**
- **Ethnographic observation**

Analysis frameworks:

- **Thematic analysis** — identifying recurring patterns across data
- **Grounded theory** — building theory inductively from the data
- **Discourse analysis** — analyzing how language constructs meaning

In qualitative theses, the method chapter describes the methodology (e.g., grounded theory), data collection (who, how many, how recruited), and data analysis procedures.

6.6. Experience sampling (ESM)

An experience sampling study asks participants to complete brief surveys repeatedly over several days or weeks — typically triggered by a timer or an event (e.g., each time they use social media). This captures behavior and experiences as they happen, rather than relying on retrospective recall.

When to use it: When your research question is about how experiences or behaviors fluctuate *within* people across situations, rather than average differences between people.

Key considerations:

- Requires a sampling app (e.g., m-Path, movisensXS) and careful pilot testing
- Participants complete many measurements — keep each survey very short (2–5 minutes)
- Analysis uses multilevel modeling (observations nested within persons)
- Ethics and burden: participants must be debriefed and compensated appropriately

Example: In a two-week ESM study (Masur, 2019), n = 164 participants completed a brief survey each time they engaged in digital communication, capturing 1,104 situational events. The analysis found that 72% of the variance in self-disclosure lay *within* persons across situations — a finding a one-time survey would have missed entirely.

6.7. The analysis plan

Your **Research Proposal must include an analysis plan** that specifies (for each hypothesis or sub-question) exactly which statistical test on which variables will be used to answer it. For example:

> *"H1 will be tested with a simple linear regression with social media exposure as the predictor and vaccine hesitancy as the outcome variable."*

This plan prevents ad-hoc analysis decisions later. In the final thesis, the analysis plan is not a separate chapter. Instead, it is integrated into the Results section.

> Pre-registration
>
> I strongly suggest to preregister your analysis plan on the Open Science Framework (OSF) before data collection. See Open Science for details. We will discuss this right before data collection.

6.8. References

- Masur, P. K. (2019). *Situational Privacy and Self-Disclosure: Communication Processes in Online Environments*. Springer International Publishing. https://doi.org/10.1007/978-3-319-78884-5
- Panizza, F., Kyrychenko, Y., & Roozenbeek, J. (2026). Survey-taking AI tools surpass human abilities. Here's what we can do about it. *Nature*. https://doi.org/10.1038/d41586-026-00386-2
- Parry, D. A., Davidson, B. I., Sewall, C. J. R., Fisher, J. T., Mieczkowski, H., & Quintana, D. S. (2021). A systematic review and meta-analysis of discrepancies between logged and self-reported digital media use. *Nature Human Behaviour*, *5*, 1535–1547. https://doi.org/10.1038/s41562-021-01117-5

7. Ethics & Data Management

7.1. The ethical self-check

Before you start collecting data, it is **mandatory** to complete the **ethical review self-check** via the Research Ethics Review Committee (RERC) of the Faculty of Social Sciences (FSW).

You can find the self-check and all relevant information here: vu.nl/en/employee/research-support/research-ethics-review-ssc

> ❗ Always complete the self-check with me
>
> I hold final responsibility for the accuracy of your answers. **Never complete the self-check without consulting me first.** Inaccurate answers can invalidate the ethical assessment and make your data unusable.

7.1.1. How the self-check works

1. Complete the online survey in consultation with me
2. If your research **meets** the faculty's ethical guidelines → no further review needed. You will receive a PDF confirmation by email.
3. If your research **does not meet** the guidelines (e.g., no informed consent, stimuli that may distress participants, or vulnerable populations) → further steps are needed

7.1.2. If your research does not pass the self-check

1. Discuss the issue with me. In many cases, a small design adjustment solves the problem
2. If the issue cannot be resolved, I will contact the RERC (not you!)
3. Note: thesis projects are generally **not eligible** for full RERC review, unless the project is part of a larger, supervised research line that will be published. In that case, I must prepare materials for full review.

7.1.3. Practical notes

- Save the self-check PDF output and store it in your surfdrive folder and attach it as an appendix to your thesis
- You can change your answers before the final submission of the form, but always discuss all answers with me before pressing submit
- In some cases, I have already applied for ethical review for a research line you are joining. Check with me whether you need to complete the self-check separately.

7.1.4. Qualtrics: disable personal data collection

By default, Qualtrics stores personal data (IP address and geolocation). If you want to keep responses anonymous:

1. Go to *Survey Options* in Qualtrics
2. Under *Survey Termination*, enable: **"Anonymize Response — Do NOT record any personal information and remove contact association"**

If you do not do this, you must answer "yes" to the question about collecting personal/sensitive data in the self-check, and you must store the data securely. We almost never need to collect IP addresses or geo location.

7.2. Data management

Careful and secure handling of research data is required throughout your thesis.

7.2.1. Storing your data (surfdrive/research drive)

The accepted secure method for storing and sharing data with me is a **shared SURFdrive or Research Drive folder**, protected by a password. I will create this folder and grant you access.

Workflow:

1. Place the **complete raw data file** in the shared folder with a clear name (e.g., `ALL_RAW_DATA_StudyName_StudentName.sav`)
2. Make a **duplicate** of the raw file — work only in the copy
3. **Remove all personal/sensitive data** from the working file (e.g., geolocation, IP addresses, names, email addresses, in case these data are still included)
4. Clean the working file and create new variables as needed
5. Keep your **syntax** (R or SPSS) updated and annotated throughout

7.2.2. Syntax hygiene

Always work with syntax — never run analyses from menus without recording them (see also the chapter "R for Data Analysis"):

- In **SPSS**: click "Paste" for every command run from a menu; add notes with `* comment text`
- In **R**: annotate your script with `# comment text`

Your syntax should allow anyone (including you, six months from now) to:

- Re-create the final dataset from the raw data
- Reproduce every reported analysis

At the start of your method section, report all cases removed from the dataset and the reasons (e.g., *"9 cases were excluded because they did not complete the questionnaire"*).

Syntax is part of your grade

Reproducible syntax is a sign of scientific rigor. I need access to your data and syntax (via surfdrive) when assessing your thesis.

8. Writing the Thesis Plan

The **MA Thesis Plan** (sometimes called the Exposé) is the first formal deliverable of your thesis project. It is due at the end of Period 4 and should be a minimum of two pages. Think of it as a working blueprint: it does not need to be polished or complete, but it should show that you have done serious preparatory thinking.

8.1. What the Thesis Plan needs to contain

8.1.1. 1. Literature overview and research gap

Begin by summarizing the state of knowledge in your topic area. You do not need an exhaustive review — focus on the core literature directly relevant to your research focus. Then clearly articulate the **research gap**: what is not yet known, understudied, inconsistent, or unresolved?

A good research gap statement follows this pattern:

> *"While prior research has shown that X, relatively little attention has been paid to Y. In particular, it remains unclear whether/how/to what extent [gap]. This matters because [theoretical/societal reason]."*

Typical length in the plan: 1–2 paragraphs.

8.1.2. 2. Research question

State your central research question explicitly. At this stage it may still be somewhat provisional — you can refine it in the Research Proposal. A good research question is:

- Specific about the constructs involved
- Empirically answerable
- Clearly connected to the gap you identified

Example:

> *"To what extent does the framing of climate change news as a political versus scientific issue affect readers' policy support, and does this effect depend on prior environmental attitudes?"*

8.1.3. 3. Hypotheses or sub-questions

List your expected hypotheses (quantitative research) or sub-questions (qualitative research). At this stage, these can be tentative. You should be able to provide at least a brief theoretical rationale for each.

For quantitative research, a hypothesis predicts a specific directional or non-directional relationship:

> *"H1: Exposure to politically framed climate news will be associated with lower policy support than scientifically framed news."*
>
> *"H2: This effect will be moderated by prior environmental attitudes, such that the framing effect is stronger among politically non-engaged readers."*

8.1.4. 4. Research design

Describe how you plan to answer your research question. Be concrete — specify the type of research and the key elements. For example:

> *"I plan to conduct a 2 (news framing: political vs. scientific) × 2 (headline valence: positive vs. negative) between-subjects online experiment. Participants will be recruited via Prolific. The main dependent variable is policy support, measured with an established 5-item scale."*

or

> *"I will use semi-structured interviews with 15–20 journalists at Dutch national news outlets to explore how they perceive and enact editorial independence norms in the context of audience metrics."*

8.2. What makes a strong Thesis Plan?

Strong	Weak
Research gap is clearly derived from the literature	Gap is vague or asserted without evidence
RQ is specific and names constructs	RQ is too broad or not researchable
Hypotheses follow logically from the theoretical argument	Hypotheses are not justified
Design is clearly matched to the RQ	Design is described only in general terms
Writing is clear and coherent	Writing is unfocused, ideas are scattered

8.3. Common mistakes

- **Describing an area of interest rather than a gap.** "Social media and mental health is an important topic" is not a gap. What specifically is unknown?
- **Overly broad RQs.** If your RQ could be a book, it's too broad. Narrow down to one specific relationship or phenomenon.
- **Missing the connection between design and RQ.** Your design must be capable of answering your RQ. If your RQ asks about causal effects, a cross-sectional survey alone is insufficient.
- **Writing the plan after the fact.** The plan is meant to guide your proposal, not just describe what you decided already. Use it to think through whether your project is feasible.

8.4. Practical tips

> Start earlier than you think
>
> The Thesis Plan is due at the end of P4, but the earlier you start reading and writing, the better your plan will be — and the easier your Research Proposal will become.

- **Use Zotero from the start**: track every paper you read with a brief note on its relevance to your project
- **Write in parallel with reading**: don't wait until you've read everything to start writing
- **Discuss your gap with me early**: I can quickly tell you if your gap has already been addressed in literature you haven't found yet
- **Keep it focused**: two pages forces you to be precise — that's the point

8.5. Template structure

A two-page Thesis Plan might be organized as follows:

```
Title (working title of the thesis)
Name | Student number | Supervisor | Date

1. Introduction and research gap (approx. ½ page)
   Brief overview of the topic; what is known; what is the gap

2. Research question (1-2 sentences)

3. Hypotheses / sub-questions (bulleted list with brief rationale)

4. Proposed research design (approx. ½ page)
   Type of study; participants/corpus; key variables/concepts; method

[Optional: preliminary reference list]
```

9. Writing the Research Proposal

The **Research Proposal** is the most important early milestone of your thesis. It is due in the first two weeks of Period 5 and is evaluated by me as *sufficient* or *insufficient*. An approved proposal is required to continue — it is the green light to start collecting data.

The proposal is not the final thesis. It is a detailed plan that shows your research design is sound, your theory is well-grounded, and you know exactly what you are going to do and how. I evaluate it primarily on **feasibility** and the **fit between your method, analysis plan, and research question**.

9.1. Required sections

9.1.1. 1. Introduction

The introduction frames your study and states your research question. It follows the same structure as the introduction in the final thesis (see Introduction & Theory Chapter), but at this stage it may not yet be in final form.

The introduction must contain:

- A clear **motivation** for the topic (why does this matter?)
- An identification of the **research gap**
- A clear, explicitly stated **research question**
- A brief argument for the **societal and scientific relevance**

Length: 1–2 pages.

Common mistakes:

- Stating why the topic is interesting in general, without identifying a specific gap
- Formulating a research question that is too broad, too vague, or not empirically answerable
- Relevance that is disconnected from the gap (e.g., claiming societal relevance for a purely theoretical contribution)

9.1.2. 2. Theory

The theory chapter reviews relevant literature and builds a theoretical argument that culminates in your hypotheses. In the proposal, this chapter should be substantially developed, though it need not be fully polished.

Structure:

1. Define and explain your central constructs
2. Review prior empirical work, organized thematically
3. Build theoretical arguments linking constructs
4. Formulate hypotheses (or sub-questions)

5. Present a **conceptual model** (for quantitative research)

Length: 4–6 pages in the proposal (6–8 in the final thesis).

Common mistakes:

- Summarizing papers one by one instead of synthesizing them thematically
- Hypotheses that appear out of nowhere without a theoretical build-up
- Inconsistent naming of constructs (use the same term throughout)
- Too many or too few hypotheses — each hypothesis needs a full theoretical argument

> The cone structure
>
> Each section of your theory chapter should start broad (definitions, background) and narrow toward your specific hypotheses. Think of it as a funnel.

9.1.3. 3. Hypotheses (or sub-questions)

State all hypotheses clearly and separately. Each hypothesis:

- Proposes exactly **one effect** (one direction, one pair of variables)
- Is numbered (H1, H2, H3 etc.; sub-hypotheses as H1a, H1b)
- Uses the **same variable names** as your theory, conceptual model, and method
- Is supported by a clear theoretical argument in the theory chapter

For **moderation**: *"The effect of X on Y will be moderated by Z, such that the effect will be stronger/weaker when Z is high."*

For **mediation**: *"The effect of X on Y will be mediated by M."*

For **qualitative research**: state sub-questions instead. Each sub-question should address one distinct aspect of the main RQ.

9.1.4. 4. Method

The method section describes how you will answer your research question. At the proposal stage, it should be as detailed as possible, even if some specifics will be refined later.

9.1.4.1. Participants / data source

- Who are the participants (or what is the corpus)?
- How will you recruit them (or sample the data)?
- What is the intended sample size? Include a **power analysis** or justify your N (e.g., for experimental research: a medium effect size with 80% power for a between-subjects ANOVA requires ~52 per cell)
- For qualitative research: how many participants, using what sampling logic?

9.1.4.2. Design

- State the type of design clearly
- For experiments: write out the full design notation (e.g., *"a 2 [X: condition A vs. condition B] × 2 [Y: high vs. low] between-subjects experiment"*)
- Name all independent, dependent, and control variables

9.1.4.3. Procedure

- Describe the study sequence step by step
- For experiments: describe exactly what each condition will look like

9.1.4.4. Measures / operationalization

For each variable:

- Name the scale or instrument
- Cite the original source
- State the number of items and the response format
- Note reliability from the original source (if available)

A first draft of the **questionnaire, experimental materials, or codebook** can be included as an appendix to the proposal.

9.1.5. 5. Analysis plan

The analysis plan is one of the most important parts of the proposal and is often where students struggle. For **each hypothesis or sub-question**, specify:

- Which statistical test you will use
- Which variables are included (exactly as named in the method section)
- What the criteria are for accepting or rejecting the hypothesis

Example analysis plan entries:

> *H1 (main effect of framing on policy support) will be tested with an independent-samples t-test, with news framing condition (political vs. scientific) as the independent variable and policy support (scale average) as the dependent variable. H1 will be supported if the effect is statistically significant ($p < .05$) and in the hypothesized direction.*
>
> *H3 (moderation of framing effect by political identity) will be tested with a 2-step hierarchical regression. In Step 1, news framing (dummy-coded) and political identity are entered as predictors of policy support. In Step 2, the interaction term (framing × political identity) is added. H3 will be supported if the interaction term in Step 2 is statistically significant.*
>
> *H5 (mediation of X on Y via M) will be tested using Hayes' PROCESS macro (Model 4) with 5,000 bootstrap samples. The indirect effect will be considered significant if the 95% bootstrap confidence interval does not include zero.*

For qualitative research: describe your analytical approach (e.g., thematic analysis following Braun & Clarke, 2006), how you will code the data, and how you will ensure trustworthiness (e.g., member checking, peer debriefing).

> **!** The analysis plan is binding
>
> I approve your analysis plan. Deviating from it substantially in the final thesis without justification is a problem. If circumstances require a change (e.g., your manipulation fails), discuss this with me immediately.

9.2. Getting to "sufficient"

I evaluate the proposal primarily on:

Criterion	What I am looking for
Feasibility	Can you realistically do this study within the timeframe?
Fit between RQ and design	Does the design actually answer the research question?
Theoretical grounding	Are the hypotheses derived from solid theoretical argumentation?
Analysis plan clarity	Do you know exactly which test to run for each hypothesis?
Consistency	Are the same variable names used throughout theory, method, and analysis plan?

9.3. Practical tips

- **Don't wait for the perfect idea.** Submit the best proposal you can. The proposal is a plan, not a guarantee. It can be refined.
- **Be very specific about your design.** Vague descriptions of "a survey" or "an experiment" are not enough. What exact manipulation? What exact measures?
- **Name your scales.** "I will measure anxiety" is not sufficient. "I will measure anxiety using the 6-item STAI-6 (Marteau & Bekker, 1992)" is.
- **Map your analysis plan against your hypotheses one by one.** If you have 5 hypotheses, you need 5 analysis plan entries — no more, no less.
- **Process peer feedback before submitting.** Fellow students can often spot inconsistencies in variable names or gaps in the theoretical argument that you miss when reading your own work.

Part III.

Writing Your Thesis

10. Structure & Formalia

Your thesis is an academic piece of work. There are many rules for how academic works should look like. Even though some of these may be arbitrary they actually serve the purpose of making different works comparable. Following these formalia is thus extremely important.

10.1. Length and formatting

	Requirement
Minimum length	8,000 words
Maximum length	9,000 words (10% margin allowed = up to 9,900)
Word count excludes	Title page, abstract, references, tables, appendices, and (for qualitative research) citations from data
Line spacing	1.5
Font	12pt Times New Roman
Pages (approx.)	22–28
Word count location	State on title page

ℹ Length is about quality, not padding

If your project is genuinely complex and cannot be reported within 9,000 words, discuss this with me. However, being overly long without justification may negatively affect your grade. Less-important material can usually be moved to an appendix.

10.2. Required structure

Your thesis must follow this structure. The page counts below are guidelines, not strict requirements.

#	Section	Guideline
0	**Title page**	—
1	**Table of contents**	—
2	**Abstract / Summary**	~0.5 page
3	**Introduction**	1–2 pages
4	**Theory**	6–8 pages
5	**Method**	2–3 pages
6	**Results**	3–4 pages (qualitative: 6–8 pages)
7	**Conclusion and Discussion**	4–5 pages

#	Section	Guideline
8	**Reference list**	30 references
9	**Statement about Generative AI and AI-Assisted Technology usage**	After references; not in word count
10	**Appendices**	—

Depending on your research design, you may deviate from this structure — but always consult me first.

10.2.1. Title page

The title page should include at minimum:

- Title of your thesis
- Your name and student number
- Course name and code (S_MTcw)
- Supervisor's name
- Date of submission
- Word count of the main text
- VU Amsterdam / Faculty of Social Sciences / Department of Communication Science

10.2.2. Abstract

The abstract (approximately half a page) is written **last** but appears first. It concisely describes:

- The reason for the study and the research question/hypotheses
- The method
- The key results
- The main conclusion

10.2.3. Appendices

At minimum, include:

- Your questionnaire and/or stimulus materials
- Your codebook (for content analysis)
- The RERC ethical self-check PDF

Variable names in the appendices should match exactly those used in your Method and Results sections.

10.3. Formatting (APA)

All text, tables, figures, the title page, table of contents, abstract, and reference list must follow **APA 7th edition** guidelines. See Citing & Referencing for details and apastyle.apa.org for the full style guide.

Key formatting rules:

- Statistical symbols are italicized: *M*, *SD*, *F*, *t*, *p*, *r*, *N*, *n*
- All numbers rounded to **two decimal places** (except $p < .001$)
- No leading zero on p values and correlations: write $p = .03$, not $p = 0.03$
- Report exact p values: $p = .03$, except when $p < .001$
- Spaces after = and < signs
- Tables and figures must include all necessary information and labels

10.4. AI disclosure statement

All theses must include a **Statement about Generative AI and AI-Assisted Technology usage** after the reference list (not counted in the word count). See Using AI Tools for what this statement must contain.

10.5. Submission

Submit your final thesis:

1. **On Canvas** (where it will be checked for plagiarism)
2. **By email** to both me and the second reviewer (check with me whether a paper version is also required)
3. **Ensure I have access** to your final data file and syntax via the shared surfdrive folder

After receiving a passing grade, you must also **upload your thesis to the UBVU thesis database** — this is a prerequisite for receiving your diploma. See Assessment & Graduation for details.

11. Introduction & Theory Chapter

11.1. The Abstract

Although it appears first in your thesis, the abstract should be the **last section you write**. It concisely summarizes:

- The reason for the study and the research question or hypotheses
- The method used
- The key findings
- The main conclusion

Keep it to approximately half a page. Focus on the most important points and do not include citations.

> The Perfect Abstract
>
> Check out this resource for how to write a top-notch abstract: https://www.nature.com/documents/nature-summary-paragraph.pdf

11.2. The Introduction

The introduction introduces the reader to your thesis. It presents your topic and frames the study by providing its scientific and societal relevance. A good introduction is **concise** and focuses on two questions: *"What has been done in this area?"* and *"Why is the current study useful?"*

11.2.1. Standard structure

1. **Reason** (*aanleiding*): Why is this topic relevant now? What social or scientific development motivates this study?
2. **Topic introduction and knowledge gap**: Briefly introduce the topic and name the gap in existing research (keep it short here; details follow in the Theory chapter)
3. **Problem statement**: Articulate the specific research problem
4. **Research question**: State the central RQ explicitly and clearly
5. **Societal and scientific relevance**: Explain why answering this RQ matters

For **qualitative research**, also briefly summarize your methodology (e.g., grounded theory) and methods (e.g., interviews, thematic analysis) at the end of the introduction.

11.2.2. Example introduction

Mattis and colleagues (2025, Journal of Communication) begin their article on nudges in news recommenders with a statement on societal relevance:

> In recent years, there has been a growing interest in the notion of "democratic news recommender design" (Helberger, 2019; Mattis et al., 2024b; Mitova et al., 2023; Vrijenhoek et al., 2021). Drawing on nudge theory (Thaler & Sunstein, 2009), this line of research posits that news recommender systems can be viewed as choice architectures, where decisions on how to select, rank, and present news items can have implications for how people engage with and process news content (Knudsen, 2023; Mattis et al., 2024b). This may have positive downstream effects on democratically relevant outcomes, such as citizens' political learning, political participation, or tolerance (Helberger & Wojcieszak, 2018; Mattis et al., 2024b).

Their argument is basically that news recommenders needs to be better understood because they can influence how people learn about the world and thus participate in democracies.

They then move on to outline the research gap:

> As of now, work on democratic news recommender design is still in its infancy, as many studies predominantly focus on conceptualizing potential approaches (Lorenz-Spreen et al., 2020; Mattis et al., 2024b; Vrijenhoek et al., 2021) and their ethical implications (Vermeulen, 2022). These studies often point to the potential of diversity-aware recommender design or nudging, but rarely test these assumptions (for exceptions see Heitz et al., 2022, 2023; Mattis et al., 2024a). Moreover, while the effects of democratic news recommender design are likely to transpire over longer periods of time, there are still rather few longitudinal research studies (for long-term studies that are an exception to this, see Heitz et al., 2022, 2023; Lee & Suh, 2022). Hence, we still know little about how nudging news engagement works over extended periods of time.

They denote that experimental work in this area, particularly over longer periods of time, are missing.

Last, they explain how their present work addresses this app, while simultaneously summarizing the research design already:

> To provide more realistic insights into the promises of democratic news recommender design, we conducted a preregistered seven-day long field experiment with Informfully (Heitz et al., 2024a),1 a fully customizable content delivery platform that we used to mimic a news aggregator app; Informfully can be downloaded directly from Google Play and the iOS app store and provided respondents with a realistic news use experience, thereby maximizing external validity. Within this externally valid news aggregator app, we tested how a position nudge and accessibility nudge affect (a) the selection of, (b) the engagement with, and (c) the learning from environmental news. The position nudge denotes the deliberate prominent positioning of a particular article, whereas the accessibility nudge describes changes to an article that make its content easier to understand by lowering its syntactic text complexity.

11.2.3. Checklist for the introduction

- ☐ Is a clear motivation for the research problem provided?
- ☐ Is the research problem clearly delineated?
- ☐ Is the research gap clearly identified?
- ☐ Are central concepts clearly defined and consistently named throughout the thesis?
- ☐ Is the research question stated explicitly and clearly?
- ☐ Is the research question researchable?
- ☐ Is both the **societal** and **scientific** relevance demonstrated?

11.3. The Theory Chapter

The theory chapter reviews the relevant literature and major theories, shows how your research builds on existing work, and leads to your hypotheses (or sub-questions for qualitative research). It is of utmost importance that it guides the reader towards the key goal (i.e., testing the hypotheses, answering the research questions). It should represent a logical rationale, not just a mix of found studies and summaries of their findings.

11.3.1. Structure

A theory chapter typically:

1. Defines and explains the central constructs/concepts
2. Builds a theoretical argument for the relationships between constructs
3. Discusses relevant prior empirical findings
4. Combines both theory and empirical findings into a coherent rationale for your study
5. Culminates in clearly formulated hypotheses (or sub-questions) and a conceptual model

Each sub-section should follow a **cone shape**: start with general definitions and background, then narrow to specific relationships relevant to your hypotheses.

> The golden thread
>
> Your research question should be "visible" in every paragraph of your thesis. Whenever you are unsure whether a paragraph belongs, ask yourself: *does this contribute to answering my research question?* If not, it probably should not be included!

11.3.2. Example theory section

At the end of the first part of their paper, Schemer et al.. (2021) write e.g., the following:

> Taken together, previous studies are inconclusive with respect to the overall relationship between the frequency of Internet or SNS use and subjective well-being. Some Internet applications or modes of use may be beneficial while others may be detrimental. A look at reviews and meta-analysis that summarized the primarily cross-sectional evidence suggests that studies documenting negative relationships outweigh those that found positive relationships (Huang, 2010; Steers, 2015). This finding may reflect that passive use of Internet applications is dominant (Escobar-Viera et al., 2018; Verduyn et al., 2015) or that processes that are detrimental to

> subjective well-being (negative feedback, upward comparisons, displacement of offline relationships) occur more frequently than processes that produce positive outcomes (positive feedback, building social capital, engage in identity exploration) when using Internet application like SNS (de Vries & Kühne, 2015; Haferkamp & Krämer, 2011).

From this, they deduce the following hypothesis in this way:

> Facing these negative net effects, the present study starts from the following between-person hypothesis (H1): Frequency of: (a) Internet use in general; and (b) SNS use in particular are negatively related to subjective well-being (between-person correlation).

> ⚠ Not just a summary of existing literature
>
> A good theory section present theoretical arguments that are supported by the existing literature and thus serves to rationalize the hypotheses. It is not a unstructured list of some related findings!

11.3.3. Content checklist

- ☐ Is the theory relevant to the research question?
- ☐ Is the literature up-to-date (primarily recent peer-reviewed journal articles)?
- ☐ Is a critical stance taken (noting relationships, contradictions, gaps)?
- ☐ Are highly relevant studies discussed in appropriate detail (including their methods and findings)?
- ☐ Is the labeling of variables/constructs consistent throughout?

11.3.4. Structure and writing checklist

- ☐ Is there a clear logical thread (*rode lijn*) throughout?
- ☐ Does each section begin with a topical sentence?
- ☐ Are sections balanced in length?
- ☐ Is the conceptual model (for quantitative research) included at the end?
- ☐ Is every construct in the hypotheses represented in the conceptual model?

11.3.5. Vocabulary and references

- ☐ Is the vocabulary academic (avoiding contractions and colloquialisms)?
- ☐ Is every claim supported by a reference?
- ☐ Are in-text citations in correct APA format?
- ☐ Are direct quotations justified, and do they include a page number?

11.3.6. Hypotheses checklist

- ☐ Does each hypothesis propose one effect only (not multiple)?
- ☐ Are moderation and mediation effects in separate hypotheses?
- ☐ Are terms used consistently across theory, conceptual model, and method?
- ☐ Is each hypothesis supported by 3–5 prior studies?

11.4. References

- Mattis, N., Heitz, L., Masur, P. K., Moeller, J., & van Atteveldt, W. (2025). Nudges for news recommenders: Prominent article positioning increases selection, engagement, and recall of environmental news, but reducing complexity does not. Journal of Communication. https://doi.org/10.1093/joc/jqaf019
- Schemer, C., Masur, P. K., Geiss, S., Müller, P. & Schäfer, S. (2021). The impact of Internet and Social Media Use on Well-Being: A Longitudinal Analysis of Adolescents Across Nine Years. Journal of Computer-Mediated Communication, 26(1),1–21. https://doi.org/10.1093/jcmc/zmaa014

12. Method Chapter

The method chapter is written in **past tense** and must be detailed enough that another researcher could fully **replicate your study**. It is highly standardized in structure.

> Language: Past tense
>
> Descriptions of *your* findings in the result section → **past tense** (*"The results showed that..."*)

12.1. Quantitative research (survey and experiment)

12.1.1. Sample and Participants

- Method of **recruitment** (convenience sampling, snowballing - How were participants recruited?)
- **Sample size**, including justification (a priori power analysis)
- **Exclusions**: exactly how many participants were excluded and for what reason(s)
- The **final** N (must match the N reported in the results section)
- **Demographics**: frequency and percentage for gender, age, education, etc.

Example:

> *"Participants were recruited via Prolific Academic. A power analysis based on an expected small-to-medium effect (r = .20, = .05, power = .80) indicated a required sample of N = 197. A total of 231 participants were recruited. After excluding 19 participants who failed two or more attention checks and 12 who completed the survey in under four minutes, the final sample consisted of N = 200 participants (54% female; M_{age} = 29.8, SD = 7.6, range: 18–61; 51% with a university degree)."*

12.1.2. Design

- State the **type of design** clearly: survey, experiment, between-subjects, within-subjects, mixed
- Describe the **task** if there was one (e.g., *"Participants were exposed to a news article and then completed a questionnaire"*)
- Name the **independent and dependent variables**
- For experiments: describe factors and conditions clearly using design notation (e.g., *"a 2 [logo color: red, blue] × 2 [medium: TV, newspaper] between-subjects design"*), and confirm randomization

12.1.3. Procedure

Describe in **chronological order**:

- What participants were exposed to (especially condition differences in experiments)
- What tasks and measurements were completed
- Key instructions given (purpose of study, debriefing)

12.1.4. Materials

- Describe **experimental stimuli** or materials, including exact differences between conditions
- All materials must be provided in the appendix (or even in the text itself if it helps the reader)

12.1.5. Measures

For each variable, report:

- Variable name (in **bold** or *italics*)
- What it measures
- Source reference (original scale)
- Number of items
- Example item(s) — or all items if few; refer to appendix for longer scales
- Response format (e.g., *"5-point Likert scale from 1 = strongly disagree to 5 = strongly agree"*)
- *M* and *SD* in your sample, along with minimum and maximum values
- Reliability (Cronbach's and/or McDonald's omega), but also results of confirmatory factor analyses if applicable!
- How the final score was computed (e.g., average of items; recoded items)
- Direction (e.g., *"Higher scores indicate greater vaccine hesitancy"*)
- For experiments: a **manipulation check**!

Eventually organize measures under subheadings: Independent variables, Dependent variables, Control variables.

Example:

> ***Privacy concerns*** *were measured with four items adapted from Malhotra et al. (2004; e.g., "I am concerned about threats to my personal privacy online"). Items were answered on a 5-point Likert scale (1 = strongly disagree; 5 = strongly agree). Confirmatory factor analysis confirmed a single-factor structure (CFI = .97, RMSEA = .05). The scale showed good reliability (= .84; M = 3.12, SD = 0.91, range: 1–5). Higher scores indicate greater privacy concerns. The final score was computed as the mean of all four items.*

Consistency is critical

The variable names in your method section must be **identical** to those used in your hypotheses, conceptual model, and results section. This is one of the most common sources of errors in MA theses.

12.2. Content analysis

The method chapter for a content analysis includes:

- A short description of the **research design**
- **Data collection**: how the corpus was assembled (sampling strategy, date range, sources)
- **Content analysis method**: manual (intercoder reliability) or automated (tool used, e.g., R, AmCat, Coosto, LLMs)
- **Operationalization**: search terms or codebook
- **Reliability**: intercoder agreement or precision/recall in relation to a gold standard

12.3. Qualitative research

The method chapter for qualitative research describes:

- **Methodology**: the chosen framework (e.g., grounded theory) and its rationale
- **Case study** (if applicable): why this case is appropriate for the research question
- **Participants/key informants**: recruitment method, demographics, sample size, inclusion/exclusion criteria
- **Data collection**: method (interviews, observation, etc.), materials used (topic guide, prompts), where conducted
- **Data analysis**: analytical approach (thematic analysis, grounded theory coding, discourse analysis), analytical materials

All choices should be supported by methodological literature with proper APA citations.

12.4. Overall checklist

- ☐ Is the method section written in past tense?
- ☐ Is the level of detail sufficient for replication?
- ☐ Is there a clear fit between the method and the hypotheses/sub-questions?

13. Results Chapter

The results section is a **matter-of-fact description** of your findings. It is written in **past tense** and focuses on answering your research questions and testing your hypotheses. Extensive interpretation belongs in the Discussion. Here you only report what you found.

That said, good results sections are well-structured and nonetheless tell a coherent "story" that guides the reader through the analyses step by step. It should usually align with the order of the hypotheses or research questions.

> Do not interpret in the Results section
>
> Reporting a finding (*"H1 was supported"*) is fine. Explaining *why* it occurred, what it means theoretically, or how it relates to prior literature is **not** — save that for the Discussion.

> Language: Past tense
>
> Descriptions of *your* findings in the result section → **past tense** (*"The results showed that…"*)

13.1. Structure for quantitative research

Report sections in this order:

13.1.1. 1. Descriptive statistics

Report means (M), standard deviations (SD), and correlations among measured variables. Present these in a table or figure. In the text, describe the main and noteworthy correlations. Note whether control variables correlate with (in)dependent variables. If so, you may want to include them in follow-up analyses.

Missing data: Briefly state how missing data were handled (e.g., listwise deletion, multiple imputation) and how many cases were excluded or imputed. Reviewers will expect this.

Scale reliability: For each multi-item scale, report internal consistency using Cronbach's or McDonald's (preferred in APA 7). This can appear in the descriptive statistics table or in a brief sentence in text.

Example:

> *"All scales showed acceptable reliability: social media use (= .84), privacy concern (= .79), and self-disclosure intentions (= .91)."*

13.1.2. 2. Manipulation check (for experiments)

Verify that the experimental manipulation worked as intended. Also check randomization: are demographic variables (gender, age, education) balanced across conditions? Report these checks even if no significant differences were found.

Example:

> *"As a manipulation check, participants in the high-exposure condition reported significantly more perceived peer disclosure than those in the low-exposure condition (M = 4.12, SD = 0.73 vs. M = 2.31, SD = 0.68), t(198) = 17.34, p < .001, d = 2.43, confirming that the manipulation was successful. Randomization checks showed no significant differences between conditions in age, t(198) = 0.84, p = .402, gender distribution, ^{2}(1) = 0.62, p = .431, or education level, ^{2}(2) = 1.14, p = .565."*

13.1.3. 3. Hypothesis testing (confirmatory analyses)

For each hypothesis:

1. **Restate** the hypothesis briefly
2. **Name the test** used and which variables are included (e.g., *"We conducted a linear regression with X as predictor and Y as outcome"*)
3. **Report results** with all relevant statistics
4. **State a brief conclusion** (e.g., *"H1 was supported"* or *"H1 was not supported"*)

Example (all four steps):

> *H1 predicted that higher exposure to peer disclosures would increase participants' own disclosure intentions. We tested this using a simple linear regression with peer disclosure exposure as the predictor and disclosure intentions as the outcome. The analysis revealed a significant positive effect, b = 0.34, SE = 0.08, 95% CI [0.18, 0.50], t(298) = 4.25, p < .001, = .24. H1 was supported.*

Follow the same order as in the Theory chapter. Also briefly report non-significant findings — do not omit them.

13.1.4. 3a. Reporting regression models

When reporting a multiple regression, first report the overall model, then individual predictors.

Overall model: Report R^2, adjusted R^2, *F*(df1, df2), *p*, and — if comparing nested models — ΔR^2 and its significance.

Individual predictors: For each predictor, report *b*, *SE*, 95% CI, *t*(df), *p*, and .

Example:

> *"A multiple regression was conducted with social media use and age as predictors of well-being. The model was significant, F(2, 297) = 14.32, p < .001, R^2 = .09, adjusted R^2 = .08, indicating that the predictors explained 9% of the variance in well-being. Social media use was a significant negative predictor, b = −0.31, SE = 0.07, 95% CI [−0.45, −0.17], t(297) = −4.43, p < .001, = −.28. Age was not a significant predictor, b = 0.04, SE = 0.03, 95% CI [−0.02, 0.10], t(297) = 1.33, p = .184, = .07."*

13.1.5. 3b. Mediation and moderation (if applicable)

Mediation (indirect effects): Report the indirect effect with a bootstrap confidence interval. If the CI excludes zero, mediation is supported.

Example:

> *"The indirect effect of social media use on well-being through social comparison was b = −0.12, SE = 0.04, 95% CI [−0.21, −0.04], indicating significant mediation."*

Moderation (interaction effects): Report the interaction term and, if significant, follow up with a simple slopes analysis or plot to illustrate the pattern.

Example:

> *"The interaction between social media use and gender was significant, b = 0.18, SE = 0.06, t(296) = 3.01, p = .003, = .15. Simple slopes analysis showed that the effect of social media use was significant for women (b = 0.31, p < .001) but not for men (b = 0.08, p = .21)."*

13.1.6. 3c. SEM and CFA (if applicable)

If you use **confirmatory factor analysis (CFA)** to validate your measurement model, report it before hypothesis testing. If you use **structural equation modeling (SEM)**, report model fit before presenting path coefficients.

Fit indices to report: ^{2}(df), *p*, CFI, RMSEA (with 90% CI), SRMR. Common thresholds: CFI .95, RMSEA .06, SRMR .08 — but treat these as guidelines, not hard cutoffs.

Example (CFA):

> *"A CFA confirmed the three-factor structure of the scale. Model fit was acceptable: ^{2}(84) = 112.43, p = .024, CFI = .97, RMSEA = .04, 90% CI [.01, .06], SRMR = .05. All factor loadings were significant and exceeded .50."*

Example (SEM path):

> *"The structural model showed a significant direct path from social media use to privacy concern (= .31, SE = .06, p < .001) and from privacy concern to protective behavior (= .24, SE = .07, p = .001). Model fit was good: CFI = .96, RMSEA = .05, SRMR = .06."*

13.1.7. 4. Secondary (exploratory analyses, usually optional)

If you want to explore the data further (e.g., covariate analyses with control variables), include these in a clearly labeled separate section.

13.2. APA statistical reporting

Follow APA 7th edition formatting:

- Statistical symbols are italicized: M, SD, N, n, F, t, r, p, b, , ^{2}
- Round all numbers to **two decimal places**
- No leading zero on p values, correlations, or standardized effect sizes that have an upper bound of 1: $p = .03$, $r = .45$ (not 0.03, 0.45)
- Report **exact *p* values**: $p = .003$ — except use $p < .001$ for very small values
- Put spaces after = and <
- Report **effect sizes** (Cohen's d, ^{2}, f^2, etc.)
- Report **95% confidence intervals** where applicable: $b = 0.34$, 95% CI [0.18, 0.50]

Examples:

> The regression analysis revealed a significant positive effect of social media exposure on vaccine hesitancy, $b = 0.34$, $SE = 0.08$, 95% CI [0.18, 0.50], $t(298) = 4.25$, $p < .001$, $= .24$.
>
> An independent-samples t-test showed no significant difference between conditions, $t(148) = 1.12$, $p = .264$, $d = 0.18$, 95% CI [−0.15, 0.51].
>
> The ANOVA yielded a significant main effect of logo color, $F(1, 196) = 8.34$, $p = .004$, $^2 = .04$, 95% CI [.01, .10].

! Report all effects, including non-significant ones

Non-significant findings are still findings and should also be reported fully!

13.3. Tables and figures

- Use tables and figures only when they **add information** that cannot be conveyed efficiently in text
- Format all tables in **APA style** (not copy-pasted from SPSS output)
- Every table and figure needs:
  - A title (above for tables, below for figures)
  - Clear axis labels or column headers
  - A note if abbreviations are used
- Variable labels in tables must match those used in the method section

13.4. Results in qualitative research

In qualitative results, the chapter demonstrates your analysis and presents empirical findings. The exact structure depends on your analytical approach, but the general logic is the same: organize around your **sub-questions or themes**, present your analytical interpretation, and support it with data.

Approach	Typical structure
Thematic analysis (Braun & Clarke)	Organized by themes derived from the data
Grounded theory	Organized by categories from selective coding
Discourse/content analysis	Organized by discursive patterns or code categories

For each sub-question or theme:

1. Introduce the theme or sub-question
2. Present the analytical finding (your interpretation, not just a summary)
3. Support findings with representative quotes from the data
4. Briefly conclude what this finding means for the sub-question

Quotes serve as illustrations — not as substitutes for analysis. Contextualize each quote before and after: explain what it illustrates and why it is representative. Format quotes in italics, indented. Use a respondent ID (e.g., *R04*, *P12*) rather than identifying information.

Example:

> *Participants frequently described feeling pressure to present an idealized version of themselves online. One respondent captured this tension clearly:*
>
> > *"I know it's not real, but I still spend like 20 minutes picking the right photo. You want people to think you're doing well, you know?" (R07)*
>
> *This illustrates the gap between participants' awareness of impression management norms and their continued compliance with them — a pattern present across 14 of the 18 interviews.*

How many quotes? Aim for 2–4 quotes per sub-question. Each quote should be the clearest illustration of its point — prefer fewer, well-contextualized quotes over many loosely connected ones.

On saturation and frequency: You may note how widespread a theme was (e.g., *"this pattern appeared across 12 of 18 interviews"*), but avoid reducing qualitative findings to counts. The goal is analytical depth, not frequency tallying.

Avoid discussing theoretical implications extensively in the results section — save that for the Discussion.

13.5. Results checklist

- ☐ Descriptive statistics (means, SDs, correlations) are reported in a table or figure
- ☐ For experiments: manipulation check and randomization check are included
- ☐ Each hypothesis or research question is addressed in order
- ☐ Every test names the statistical method, variables, full statistics, and a brief conclusion
- ☐ Non-significant findings are reported, not omitted
- ☐ All statistics follow APA 7 formatting (italics, no leading zeros, exact p values)
- ☐ Effect sizes are reported for every test
- ☐ Tables and figures are APA-formatted (not raw software output)
- ☐ No interpretation or theoretical discussion — that belongs in the Discussion

13.6. References

- Masur, P. K., DiFranzo, D. J., & Bazarova, N. N. (2021). Behavioral Contagion on Social Media: Effects of Social Norms, Design Interventions, and Critical Media Literacy on Self-Disclosure. PLOS One, 16(7). e0254670. https://doi.org/10.1371/journal.pone.0254670

14. Conclusion & Discussion

The conclusion and discussion chapter is the final section of your thesis. It is where you step back from the data and reflect on the meaning, implications, and limitations of your findings.

> Language: Present tense
>
> Descriptions of *your* findings → **past tense** (*"The results showed that…"*) But theoretical implications and conclusions → **present tense** (*"This suggests that…"*)

14.1. Standard structure (use max. 3 subheadings)

14.1.1. 1. Summary and conclusion

Opening paragraph:

- Restate the main focus and purpose of the study
- Briefly mention the approach (design)
- Give a clear, concise overall summary of the answer to your research question

Middle part — detailed summary and discussion of findings:

For each hypothesis or sub-question:

- Summarize the finding
- Relate it to the **specific literature** used to motivate the hypothesis
- Explain whether it is consistent or inconsistent with prior work and *why*
- For unexpected findings: introduce possible explanations, potentially drawing on additional literature or theories
- Offer multiple interpretations where relevant
- Make cross-connections between findings

The following are illustrative examples of what good discussion paragraphs look like. They are not taken from actual publications.

Example — supported hypothesis:

> *"H1 was supported: higher exposure to peer disclosures increased participants' own disclosure intentions ($b = 0.34$, $SE = 0.08$, $p < .001$, $\quad = .24$). This is consistent with the behavioral contagion hypothesis (e.g., Cialdini & Goldstein, 2004; Masur et al., 2021), which predicts that observed disclosures function as descriptive social norm signals that increase perceived pressure to disclose. The effect size ($\quad = .24$) aligns with prior work on norm-based social influence online (Lapinski & Rimal, 2005), though it is somewhat smaller than found in a controlled laboratory setting ($d = 0.44$; Masur et al., 2021), which may reflect the more naturalistic conditions of the current survey."*

Example — unexpected null result:

> *"Contrary to H3, privacy literacy was not significantly associated with protective privacy behavior ($b = 0.07$, $SE = 0.08$, $p = .38$, $= .06$). This is inconsistent with prior studies that found a positive link between privacy knowledge and protective action (Bartsch & Dienlin, 2016; Masur & Scharkow, 2020). One possible explanation is that knowledge alone is insufficient to motivate behavior change — a pattern documented in the privacy paradox literature (Acquisti et al., 2015). The null finding may also reflect limited variance in privacy literacy scores in the current student sample, which may have reduced statistical power to detect a small effect. Future research should replicate this test in more heterogeneous samples."*

Common mistakes to avoid:

- Saying *"This finding is in line with previous research (ref1, ref2)"*, also explain *how* and *why*
- Ignoring non-significant findings. Instead, discuss them and consider limitations, e.g., whether the study was adequately powered

Middle part — broader implications:

- **Theoretical implications**: What does your study contribute to theory? Does it extend, refine, or challenge existing frameworks?
- **Practical implications**: What do your findings mean for practitioners, policymakers, or society?
- Connect back to the relevance you established in the Introduction

Closing of the first section:

- Consider the broader picture — what has your study contributed to the field?
- Do the findings yield new insights?

14.1.2. 2. Strengths, limitations, and future research

Strengths:

- What methodological or conceptual strengths does your study have? (These give meaning to the limitations. There are always trade-offs)

Limitations:

- Be thorough and honest
- Address methodological limitations (e.g., sampling, measurement, generalizability)
- For qualitative research: discuss applicability of findings (situations, variations)

Future research:

- Suggestions must go beyond just *fixing the limitations*
- Propose new, theoretically interesting research questions that follow from your conclusions
- These suggestions should be grounded in your theoretical implications

14.1.3. 3. Closing paragraph / final conclusion

End with a strong, clear and short paragraph:

- Concisely restate the goal of the study
- State a clear and well-founded conclusion
- Provide a direct answer to the research question and hypotheses
- End with a strong take-home message

⚠ The discussion is the last thing readers read

A weak final paragraph is memorable for the wrong reasons. End confidently with a clear statement of what your study has contributed.

14.2. Full discussion checklist

- ☐ Does the opening clearly restate the research focus and approach?
- ☐ Does the summary use consistent terminology from the Introduction?
- ☐ Is each finding linked to the specific literature that motivated it?
- ☐ Are explanations given for both consistent *and* inconsistent findings?
- ☐ Are cross-connections made between findings?
- ☐ Are both theoretical and practical implications discussed?
- ☐ Is the limitation section thorough (not just a brief list)?
- ☐ Are future research suggestions theoretically grounded?
- ☐ Is there a strong closing sentence?

Part IV.

Academic Practices

15. Using AI Tools

(Generative) AI tools are increasingly part of the academic workflow and will continue to be implemented in even more diverse ways. The Department of Communication Science at VU Amsterdam generally permits the use of these tools, provided you use them **responsibly, ethically, and transparently**! What constitutes such use, depends entirely on the nature of your research and the specific type of AI use you are planning to implement.

> ⚠ Key problem: The stochastic parrot
>
> Current AIs, such as chatGPT, Gemini or Claude, are based on an architecture called transformers, which are - simply speaking - large prediction machines. Based on an input (e.g., a question), they deliver the most likely answer. They do not "know", "reason" or "think" in the human-sense. Their answer thus entirely depends on their training data and word similarities, which can be misleading and plain wrong. Don't be fooled: these tools may output text that sounds coherent or logical, even convincing, while being based on dubious sources or completely hallucinated!

> ❗ The core principle: Human in the Lead
>
> You are always the one who decides on the content. You are responsible for the quality and originality of your work. Generative AI cannot do the thinking for you — and it should not!

15.1. What is responsible use?

Responsible use means:

- **You decide on the content** at all times
- **You are critical of AI output** — AI is not a source of truth; it can produce inaccurate, incomplete, or superficial suggestions
- **You understand what you are doing** — use AI tools only when you understand the topic well enough to evaluate, improve, or reject what the tool suggests
- **You take full responsibility** for the final submitted version

Generative AI can be a useful sparring partner for brainstorming, gaining an overview, structuring your work, or improving text you have already written. It cannot replace the reading, thinking, and writing that constitute your learning process.

There is a deeper reason to stay actively involved in research tasks beyond academic honesty: it is how expertise develops. Searching and reading literature builds a map of what is known and unknown in your field. Cleaning and checking data builds intuition for data quality. Coding qualitative material builds understanding of your participants' experiences. These tasks may seem routine, but they are precisely how you develop the judgment needed to recognize when something is wrong — and to evaluate whether AI output is accurate (Messeri & Crockett, 2026).

Students who delegate these tasks to AI risk not developing the competence to supervise it responsibly.

The following examples illustrate what responsible and irresponsible use look like in practice:

Scenario	Assessment
You use Gemini to brainstorm search keywords, then search and read the sources yourself	**Responsible**
You ask ChatGPT "what are the main theories on social media and well-being?" and include those theories without reading the original sources	**Not responsible** — AI can hallucinate or misrepresent findings
You paste an R error into Copilot, read and understand the suggested fix, and verify it works	**Responsible**
You ask Gemini to "write the methods section based on these notes" and submit it with minor edits	**Not responsible** — you are not the author of that text
You upload your own PDFs to NotebookLM to identify sections worth reading more carefully	**Responsible**
You ask an AI to summarize papers you have not read and cite them in your theory chapter	**Not responsible** — you cannot vouch for what you have not read

15.2. Academic misconduct and fraud

- **Never copy-paste AI-generated text** into your thesis without attribution. This is plagiarism!
- Using AI without transparency is **academic fraud**
- If I doubt whether you have used AI responsibly, I will discuss this with you; persistent doubts go to the Examination Board

More information: vu.nl/en/student/examinations/academic-integrity

VU rules on generative AI: vu.nl/en/student/examinations/generative-ai-your-use-our-expectations

15.3. VU-licensed vs. non-licensed tools

15.3.1. VU-licensed tools (recommended)

Log in with your **VU email account**, then your data will not be used to train the model:

Tool	Access
Microsoft Copilot	copilot.microsoft.com — log in with VU credentials
Google Gemini	gemini.google.com — log in with VU email
NotebookLM	notebooklm.google.com — log in with VU email

15.3.2. Non-licensed tools (use at your own risk)

Tools like **ChatGPT**, **Elicit**, **QuillBot**, and others fall **outside VU's license**. When using these: - Your data may be used for model training - You are personally responsible for privacy and data protection - **Never upload personal research data** (names, email addresses, interview transcripts, survey data) - **Never upload paywalled articles** — this violates copyright

> ⚠ GDPR and data protection
>
> Never (!!!!) upload sensitive or personal data (e.g., data you collected from participants) to any AI tool, not even VU-licensed ones. This includes: names, email addresses, research data, financial records, or company details from internships or partnerships.
> THERE IS NO EXCEPTION TO THIS!

15.4. Responsible AI use by thesis phase

Thesis Phase	Responsible Use	Useful Tools
Topic selection	Brainstorming ideas, identifying research gaps, but always verify by cross-checking with relevant literature	Gemini, Elicit, Perplexity
Literature review	Finding keywords/synonyms, searching articles (although this is very limited and cannot replace a classic literature search), generating summaries (always verify based original source material)	Gemini, Elicit, Scite, Research Rabbit, NotebookLM, Undermind, Perplexity
Research design	Generating ideas, creating experimental stimuli (justify all choices yourself), cross-validate, develop items (evaluate carefully)	Gemini, Qualtrics AI
Proposal writing	Feedback on text you wrote; rephrasing, shortening, grammar	Gemini, Grammarly, Wordtune
Data collection	In some cases, automating data collection (verify quality yourself, implemented validation procedures if possible)	Gemini, Google Forms
Data analysis	Generating/debugging R code, creating visualizations (verify all output), automated content analysis…	Gemini, Orange3
Writing the thesis	Improving clarity and structure of your own text	Gemini, Grammarly, QuillBot, Wordtune

Thesis Phase	Responsible Use	Useful Tools
Revision & feedback	Improving academic style, tone, consistency	Gemini, Grammarly, Hemingway Editor, Trinka AI

Sustainability

Every query to a generative AI system requires significant computing energy and thus has a large environmental impact. Use these tools mindfully and purposefully!

15.5. The AI statement (required)

All theses must include a **Statement about Generative AI and AI-Assisted Technology usage** at the end of the thesis, after the reference list. This statement is **not counted in the word count**.

Your statement must answer:

1. **Did you use generative AI?** (yes/no — if no, the statement ends here)
2. **Which tools did you use, and how?** — for each tool, specify the purpose (e.g., *"I used Google Gemini to receive feedback on paragraph structure in the introduction and theory sections"*)
3. **How did you handle the AI-generated content?** — end with a statement taking responsibility for the final work

15.5.1. Example AI statements

> *"During the process of writing this thesis, I used Perplexity to explore the scientific literature. I always verified the information received via AI searches by reading the original sources. I also used Google Gemini to receive feedback on my own writing regarding paragraph structure and academic phrasing. The final formulations were written and checked by myself. I take full responsibility for the final submitted version."*

> *"When writing the theory sections, I used Grammarly to improve readability and language. I also used Copilot to debug errors in my R code during analyses. When using these tools, I carefully reviewed and edited the generated content. I take full responsibility for the content of this thesis."*

15.5.2. In-chapter disclosure

If you used AI for specific chapter-level tasks, also note this within the relevant chapter. For example:

- *Theory section*: used AI to generate an initial reading list on a topic
- *Method section*: used AI to generate experimental stimuli
- *Results section*: used AI to generate or refine analysis code

Keep a record of your AI prompts and outputs — you may be asked to show these.

15.6. References

- Messeri, L., & Crockett, M. J. (2026). The uncritical adoption of AI in science is alarming — we urgently need guard rails. *Nature*, *653*, 675–676. https://doi.org/10.1038/d41586-026-01557-x

16. Literature Search

A thorough and systematic literature review is foundational to your thesis. This chapter offers guidance on how to find, organize, and manage the literature for your research.

16.1. Where to search

VU Amsterdam provides access to major academic databases. Log in with your VU credentials at ub.vu.nl.

Database	Best for
Scopus	Broad coverage of social sciences, easy citation tracking
Web of Science	High-quality journals, citation analysis
PsycINFO	Psychology and communication research
Communication Abstracts	Communication-specific literature
Google Scholar	Quick searches, finding grey literature
JSTOR	Archive of humanities and social science journals

For **qualitative or content analysis** work, additional platforms like **LexisNexis** (news content) or **Coosto/ANP** (social media) may be relevant.

16.2. Search strategies

16.2.1. Keywords and Boolean operators

Construct your search systematically:

- Use **AND** to narrow results (e.g., `social media AND misinformation AND health`)
- Use **OR** to broaden results for synonyms (e.g., `vaccine hesitancy OR vaccine refusal`)
- Use **NOT** to exclude irrelevant terms
- Use **quotation marks** for exact phrases: `"filter bubble"`
- Use **truncation** with `*` for word variations: `persuad*` finds persuade, persuasion, persuaded

Keep a record of your search terms, databases used, and results counts — you may need to report this in your method section.

16.2.2. Snowballing

Once you find a central paper:

- **Forward snowballing**: who has cited this paper? (use Google Scholar or Scopus)
- **Backward snowballing**: what does this paper cite?

16.2.3. Scoping and narrowing

- Prioritize **peer-reviewed journal articles** from reputable journals
- Prefer **recent** literature (typically last 10–15 years, unless citing foundational work)
- Be selective — your Theory chapter should discuss only what directly contributes to your argument

Relevant Journals in our field are for example (check out this Communication Science Journal Tracker):

- Journal of Communication
- Journal of Computer-Mediated Communication
- Political Communication
- Science Communication
- New Media & Society
- Communication Monographs
- Social Media + Society
- Information Communication & Society
- Media Psychology
- Journal of Media Psychology
- Mobile Media & Communication
- and many more

Quality over quantity

The vast majority of your references should be peer-reviewed academic journal articles. Avoid blogs, Wikipedia, previous BA/MA theses, and PhD dissertations unless unavoidable.

16.3. Reference management: Zotero

I strongly recommend using **Zotero** (free) to manage your references:

- Install the browser extension to save references directly from databases
- Organize references into collections by topic
- Add notes to each reference summarizing relevant content
- Generate in-text citations and reference lists in APA format
- Export `.bib` files for Quarto, LaTeX, or Word integration

Write as you read

When reading a paper, immediately write down the relevant arguments, findings, and how they connect to your own study. Do not rely on being able to recall this later. This also prevents you from losing track of sources.

16.4. AI tools for literature search

AI tools can support (but not replace) your literature search:

Tool	Use
Elicit	AI-powered literature search and summarization
Perplexity	AI search engine with cited sources; good for quick overviews
Research Rabbit	Discover related papers via citation networks
Undermind	AI-assisted deep literature search
NotebookLM (VU-licensed)	Summarize and explore uploaded documents
Scite	Find papers that support or contradict a claim

> ❗ Always verify AI-generated literature
>
> AI tools can hallucinate references that do not exist or misattribute findings. **Always read the original sources yourself** and verify that citations are real and accurately represented before including them in your thesis.

See Using AI Tools for the full departmental AI policy.

17. Citing & Referencing

Your thesis must follow **APA 7th edition** throughout — for in-text citations, the reference list, tables, figures, and statistical reporting. The authoritative source is apastyle.apa.org.

17.1. In-text citations

17.1.1. Basic format

Situation	Format	Example
One author	(Author, Year)	(Masur, 2023)
Two authors	(Author & Author, Year)	(Masur & Teutsch, 2023)
Three or more authors	(First author et al., Year)	(Masur et al., 2023)
Narrative citation	Author (Year) found that…	Masur (2023) found that…
Multiple citations	Alphabetical order, separated by ;	(Jones, 2021; Masur, 2023)

17.1.2. Direct quotations

If you quote directly, you must include a **page number**:

> (Masur, 2023, p. 45)

> **i** Minimize direct quotations
>
> In scientific writing, you should paraphrase and synthesize — direct quotations are the exception, not the rule. Only quote when the exact wording is essential.

17.1.3. Paraphrasing

When you restate an idea in your own words, you still need a citation. Every factual claim, theoretical argument, or empirical finding that is not your own must be cited.

17.2. Reference list

The reference list appears at the end of the thesis, after the Conclusion and before the AI statement. Entries are in **alphabetical order by first author's last name** and use a **hanging indent**.

17.2.1. Common reference types

Journal article:

> Masur, P. K., DiFranzo, D. J., & Bazarova, N. N. (2021). Behavioral Contagion on Social Media: Effects of Social Norms, Design Interventions, and Critical Media Literacy on Self-Disclosure. PLOS One, 16(7). e0254670. https://doi.org/10.1371/journal.pone.0254670

Journal article with DOI: Include the DOI as a live hyperlink. Always include DOI when available.

Book:

> Masur, P. K. (2018). *Situational privacy and self-disclosure: Communication processes in online environments.* Cham, Switzerland: Springer

Book chapter:

> Masur, P.K., Hagendorff, T., & Trepte, S. (2023). Challenges in Studying Social Media Privacy Literacy. In S. Trepte & P. K. Masur (Eds.). The Routledge Handbook of Privacy and Social Media (pp. 110-123). New York: Routledge.

> ⚠ Reference managers often produce incomplete book chapter references
>
> Always verify book chapter references manually — check that editor names, book title, page numbers, and publisher are all included.

17.3. Quality of references

- The **large majority** of your references should be peer-reviewed journal articles
- Prefer articles from journals with a strong reputation in social sciences
- Avoid citing: Wikipedia, blogs, news articles, BA/MA theses, or PhD dissertations as primary sources for theoretical claims
- When in doubt about a source's quality, discuss it with me
- Aim to include diverse sources (not just from WEIRD countries or all-male). You can check the diversity of your sources with the "Diversity-X" tool.

17.4. Using Zotero

Zotero generates APA 7th references automatically. However, always **check the output** — automatic generation sometimes produces errors or incomplete entries, especially for: - Book chapters (often missing editors, page numbers, or publisher) - Conference papers - Preprints

17.5. Checklist: Reference list

- ☐ Every in-text citation has a corresponding entry in the reference list, and vice versa
- ☐ All entries follow APA 7th format
- ☐ Journal article titles are in sentence case (capitalize only first word, proper nouns, and first word after a colon)
- ☐ Journal names are in title case and italicized
- ☐ DOIs are included where available
- ☐ Book chapter references are complete (editors, page numbers, publisher)
- ☐ References are in alphabetical order by first author's last name

18. Open Science

Open science refers to a set of practices that make scientific research more transparent, reproducible, and accessible. Adopting open science practices in your thesis not only strengthens the quality of your work, but also aligns it with the standards increasingly expected in Communication Science. I, myself, strongly advocate for the implementation of open science practices (see this paper) and thus expect you to engage with these practices as well.

18.1. Preregistration

Preregistration means publicly documenting your research design, hypotheses, and analysis plan *before* you collect data. It:

- Prevents p-hacking and HARKing (Hypothesizing After Results are Known)
- Increases credibility of your confirmatory findings
- Distinguishes what you planned from exploratory analyses you add later

Why does this matter in practice? In a landmark study, 100 published psychology findings were independently replicated: only 36% replicated successfully, and the average effect size was *half* the magnitude of the originals (Open Science Collaboration, 2015). Many of these studies had been reported selectively — only the analyses that produced significant results made it into the paper. Preregistration closes this loophole: if you commit to your hypotheses and analysis plan before seeing the data, you cannot selectively report what looks good.

18.1.1. Where to preregister

The standard platform is the **Open Science Framework (OSF)**: osf.io

To pre-register:

1. Create a free OSF account (log in with your institutional email)
2. Create a new project for your thesis
3. Under *Registrations*, choose a registration template (e.g., AsPredicted or OSF Prereg)
4. Complete your hypotheses, design, and analysis plan
5. Submit — you receive a time-stamped, publicly accessible registration

> Discuss preregistration with me
>
> I strongly encourage you to preregister your MA thesis. I can help you how to do it. If you registered, include the OSF link in your method section.

18.1.2. What a preregistration looks like

A preregistration is more than a brief list of hypotheses — it should contain enough detail that someone else could run the study from it. Below is a condensed example for a survey study on privacy literacy and self-disclosure:

---

1. Hypotheses and theoretical rationale

H1: Privacy literacy will be negatively associated with self-disclosure depth on social media. Research consistently shows that privacy knowledge shapes how people evaluate disclosure risks (Masur, 2020). We therefore expect that users with higher privacy literacy will share less personal information on social media.

H2: The negative association between privacy literacy and self-disclosure will be stronger for users who also report high privacy concerns (moderation). Privacy concerns may serve as a motivational bridge between knowledge and behavior (Dienlin & Metzger, 2016). Without a concern for privacy, knowledge alone may not translate into reduced disclosure.

2. Study design, sample, and power analysis

Design: Cross-sectional online survey.

Recruitment: Participants will be recruited via Prolific Academic. Inclusion criteria: 18+ years old, active social media user (at least once per week), residing in the Netherlands.

Sample size: A power analysis for a small-to-medium correlation ($r = .20$, $= .05$, power $= .80$) indicates $N = 197$. We will recruit $N = 220$ to account for expected exclusions (attention check failures, incomplete responses). We will exclude participants who fail two or more of three attention checks.

3. Measures

Privacy literacy will be measured with the 8-item Online Privacy Literacy Scale (OPLIS; Masur & Scharkow, 2020; e.g., "Websites are legally required to have a privacy policy"). Response format: true/false/don't know. Higher scores indicate greater privacy knowledge.

Self-disclosure depth will be measured with four items adapted from Bazarova & Choi (2014; e.g., "When I post on social media, I share personal thoughts and feelings"). Responses on a 5-point scale (1 = strongly disagree; 5 = strongly agree).

Privacy concerns will be measured with the 3-item scale by Malhotra et al. (2004; e.g., "I am concerned about threats to my personal privacy online"). Responses on a 5-point scale.

4. Analysis plan

Confirmatory analyses: H1 will be tested with a simple linear regression with OPLIS score as predictor and mean self-disclosure score as outcome (controlling for age and gender). H2 will be tested by adding the OPLIS × privacy concerns interaction term to the regression model. Significant interactions will be probed at ±1 SD.

Exploratory analyses: We will additionally examine whether the associations differ by platform (Instagram vs. Facebook). These analyses are not pre-specified and will be clearly labeled as exploratory in the thesis.

18.2. Open data

Sharing your data and analysis code allows others to verify, reuse, and build on your work.

In your VU thesis, you share data with me via a secure surfdrive folder (see Ethics & Data Management). After your thesis is graded, you may optionally deposit anonymized data in a public repository such as:

- **OSF** (osf.io) — general purpose, mostly likely the best choice for your thesis
- **DANS / Data Archiving and Networked Services** (dans.knaw.nl) — Dutch research data archive
- **Zenodo** (zenodo.org) — CERN-hosted, general purpose

⚠ Never share personal data publicly

Before depositing any data publicly, ensure all personal and sensitive information has been removed. When in doubt, consult me and the RERC guidelines.

18.3. Reproducible analysis

Reproducibility means your results can be re-created exactly from your raw data and code. Good practices:

- Keep a **clean, annotated syntax file** (R or SPSS) from the start (see chapter "R for Data Analysis")
- Use **relative file paths** so your code works on other machines
- Document every data transformation and exclusion decision
- Consider using **R Markdown** or **Quarto** to combine analysis and write-up

For R users: resources are available on the Canvas page *R for Social Scientists*.

18.4. Transparency in reporting

- Report **all tested hypotheses**, including non-significant findings
- Clearly distinguish **confirmatory** (pre-registered) from **exploratory** analyses
- If you deviated from your pre-registered plan, explain why
- Report exact p values and effect sizes (not just $p < .05$)

> **Going further: Specification curve analysis**
>
> If you want to go beyond the minimum, **specification curve analysis** (Simonsohn et al., 2020; see also the `specr` package for R) is an advanced technique for showing how robust your findings are to different analytical choices — for example, different operationalizations of a construct, different sets of control variables, or different sample exclusion rules. Rather than picking one "best" specification, you run all defensible ones and plot the full distribution of results. This makes your analytical decisions transparent and guards against the impression that your findings depend on one convenient choice. It is not required for an MA thesis, but it is a sign of methodological sophistication and increasingly valued in Communication Science.

18.5. Open access

After passing your thesis, you are required to upload your thesis to the **UBVU thesis database** with public access rights (see Assessment & Graduation). This makes your work openly available for future researchers and students.

19. R for Data Analysis

R is the primary statistical computing environment used in Communication Science research at VU Amsterdam. This chapter points you to resources for learning R, running common analyses, and reporting results in APA format. It also covers how AI tools can help you write and debug R code.

19.1. Getting started with R

If you are new to R or need a refresher, start here:

Resource	What it covers
R for Social Scientists (Canvas)	The departmental resource — covers common analyses relevant to Communication Science; start here
CCS Amsterdam R Course Material	Modular R tutorials developed at VU Amsterdam for Communication Science students — covers R fundamentals, tidyverse, ggplot2, regression, multilevel models, factor analysis, text analysis, topic modeling, and more
R for Data Science (Wickham et al.)	Free online book; covers data import, transformation, visualization with the tidyverse
Learning Statistics with R (Navarro)	Free textbook; covers statistics concepts alongside R implementation
swirl	Interactive R tutorials you run inside R itself
Posit Cheatsheets	One-page visual references for ggplot2, dplyr, tidyr, etc.

Statistics helpdesk

VU Amsterdam runs a **statistics helpdesk** for students. Check Canvas for the current schedule and how to book an appointment. This is the right place for questions about which test to use and how to run it — not me.

19.2. Essential packages

Install these packages to cover most analyses you'll need:

```
install.packages(c(
  "tidyverse",   # data manipulation and visualization (dplyr, ggplot2, tidyr, ...)
  "haven",       # import SPSS (.sav) and Stata files
  "psych",       # reliability (alpha), descriptive stats, factor analysis
  "car",         # ANOVA, Levene's test, recoding
```

```
  "lme4",         # multilevel / mixed models
  "lavaan",       # SEM and mediation/moderation (alternative to PROCESS)
  "mediation",    # causal mediation analysis
  "effectsize",   # Cohen's d, eta-squared, omega-squared
  "apaTables",    # export APA-formatted tables to Word
  "papaja",       # APA-style reporting in R Markdown / Quarto
  "report"        # automatic plain-language results reporting
))
```

19.3. Common analyses

19.3.1. Descriptive statistics and correlations

```
library(tidyverse)
library(psych)

# Descriptive statistics
describe(data[, c("var1", "var2", "var3")])

# Correlation matrix (APA-ready)
library(apaTables)
apa.cor.table(data[, c("var1", "var2", "var3")],
              filename = "Table1_correlations.doc")
```

19.3.2. Reliability (Cronbach's)

```
library(psych)

# Select items belonging to one scale
alpha(data[, c("item1", "item2", "item3", "item4")])
# Look at: raw_alpha (Cronbach's  )
```

19.3.3. Independent samples *t*-test

```
# Test whether group means differ
t.test(outcome ~ group, data = data, var.equal = TRUE)

# Effect size
library(effectsize)
cohens_d(outcome ~ group, data = data)
```

Report as: $t(\text{df}) = \text{X.XX}$, $p = .\text{XXX}$, $d = \text{X.XX}$

19.3.4. One-way and factorial ANOVA

```
library(car)

# Factorial ANOVA (e.g., 2x2 between-subjects experiment)
model <- lm(outcome ~ factor_a * factor_b, data = data)
Anova(model, type = "III")   # Type III sums of squares

# Effect size
library(effectsize)
eta_squared(model, partial = TRUE)
```

Report as: F(df1, df2) = X.XX, p = .XXX, ^{2}_p = .XX

19.3.5. Linear regression

```
# Simple regression
model1 <- lm(outcome ~ predictor, data = data)
summary(model1)

# Multiple regression
model2 <- lm(outcome ~ predictor1 + predictor2 + covariate, data = data)
summary(model2)

# Standardized coefficients
library(effectsize)
standardize_parameters(model2)
```

Report as: b = X.XX, SE = X.XX, 95% CI [XX, XX], = X.XX, p = .XXX

For more comprehensive tutorials, see here.

19.3.6. Moderation (interaction effects)

```
# Center continuous predictors before creating interaction term
data$pred_c      <- scale(data$predictor, center = TRUE, scale = FALSE)
data$moderator_c <- scale(data$moderator, center = TRUE, scale = FALSE)

model_mod <- lm(outcome ~ pred_c * moderator_c, data = data)
summary(model_mod)

# Visualize interaction with ggplot2 or the 'interactions' package
library(interactions)
interact_plot(model_mod, pred = pred_c, modx = moderator_c)
```

19.3.7. Mediation analysis

```
# Using the 'mediation' package (bootstrapped indirect effects)
library(mediation)

# Step 1: model for mediator
m_model <- lm(mediator ~ predictor, data = data)

# Step 2: model for outcome
y_model <- lm(outcome ~ predictor + mediator, data = data)

# Bootstrap mediation
med_out <- mediate(m_model, y_model,
                   treat = "predictor", mediator = "mediator",
                   boot = TRUE, sims = 5000)
summary(med_out)
```

Alternatively, you can also use **lavaan** for structural equation modeling in R to run similar analyses (see this tutorial)

19.3.8. Manipuation check (experiment)

```
# Did participants in different conditions perceive the manipulation differently?
t.test(manipulation_check ~ condition, data = data)

# For 2x2 design
aov_result <- aov(manipulation_check ~ factor_a * factor_b, data = data)
summary(aov_result)
```

19.4. APA-formatted reporting in R

19.4.1. The `report` package

```
library(report)

model <- lm(outcome ~ predictor, data = data)
report(model)
# Returns a plain-language sentence you can adapt for your results section
```

19.4.2. The `apaTables` package

```
library(apaTables)

# APA regression table exported to Word
apa.reg.table(model1, model2,
              filename = "Table2_regression.doc")
```

19.4.3. papaja (APA manuscripts in Quarto/R Markdown)

If you are writing your thesis in R Markdown or Quarto, the papaja package provides APA-formatted output. Use `apa_print()` to inline statistics:

```
library(papaja)
result <- t.test(outcome ~ group, data = data)
apa_print(result)$statistic   # → "t(48) = 2.34, p = .023"
```

19.4.4. Visualizations with ggplot2

`ggplot2` (part of the tidyverse) is the standard tool for creating publication-quality figures in R. It builds plots in layers using a consistent grammar: data → aesthetics → geometric objects.

```
library(ggplot2)

# Scatter plot with a regression line
ggplot(data, aes(x = predictor, y = outcome)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(title = "Effect of X on Y",
       x = "Predictor", y = "Outcome") +
  theme_minimal()

# Bar chart of group means with error bars
data |>
  group_by(condition) |>
  summarise(mean = mean(outcome), se = sd(outcome) / sqrt(n())) |>
  ggplot(aes(x = condition, y = mean, fill = condition)) +
  geom_col(width = 0.5) +
  geom_errorbar(aes(ymin = mean - se, ymax = mean + se), width = 0.2) +
  labs(x = "Condition", y = "Mean outcome") +
  theme_minimal() +
  theme(legend.position = "none")
```

For a comprehensive step-by-step tutorial on data visualization with ggplot2, see the CCS Amsterdam visualization tutorial.

19.5. Keeping your syntax clean

Good syntax habits matter — I need to be able to reproduce your analyses from your raw data. You can use both a standard R file (.r) or an RMarkdown file (.rmd).

```
# ============================================================
# MA Thesis Analysis - [Your Name] - [Date]
# ============================================================

# 0. Load packages ----
library(tidyverse)
library(psych)
library(effectsize)

# 1. Import data ----
data_raw <- haven::read_sav("data/raw_data.sav")

# 2. Data cleaning ----
# Remove incomplete responses (n = 9)
data_clean <- data_raw |>
  filter(progress == 100)

# 3. Compute scale scores ----
# Scale: Vaccine Hesitancy (  = .82 in original; check below)
data_clean <- data_clean |>
  mutate(vacc_hes = rowMeans(across(c(vh1, vh2, vh3, vh4, vh5)),
                             na.rm = TRUE))

# 4. Descriptive statistics ----
describe(data_clean[, c("vacc_hes", "soc_media_exp", "age")])

# 5. H1: Regression of social media exposure on vaccine hesitancy ----
h1_model <- lm(vacc_hes ~ soc_media_exp, data = data_clean)
summary(h1_model)
```

Use section headers in your script

The `# Section ----` pattern (four dashes) creates collapsible sections in RStudio's code outline. This keeps your syntax file navigable when it grows long.

19.6. Getting help with R code

Beyond the statistics helpdesk and Canvas resources, AI tools can be genuinely useful for R:

Task	Useful approach
Debugging an error message	Paste the error + relevant code into Gemini or Copilot and ask what is wrong
Finding the right function	Describe what you want to do in plain English
Writing boilerplate (e.g., a for-loop)	Ask AI to generate it, then check you understand it
Interpreting output	Ask AI to explain what a specific output means

> **Verify AI-generated R code**
>
> AI tools can produce plausible-looking but incorrect R code. Always run it, check the output makes sense, and make sure you understand what every line does before including results in your thesis. See Using AI Tools for the full policy.

Part V.

Submission & Graduation

20. Submission & Assessment

20.1. Submitting your final thesis

Submit your final thesis **no later than the first-opportunity deadline** (see Timeline & Deadlines):

1. **Upload on Canvas** (required — the thesis is checked for plagiarism here)
2. **Email to me and the second reviewer**
3. **Ensure I have access** to your final data file and syntax via the shared surfdrive folder

20.2. How your thesis is assessed

Your thesis is evaluated by **two reviewers**: me (first reviewer) and a second reviewer. Both use the official **assessment form** (available on Canvas).

	Details
Time to grade	Two weeks (10 working days) after submission deadline
Final grade	Average of first and second reviewer
Disagreement	If assessors differ by more than 2 points, or one fails and the other passes → third assessor assigned by the Examination Board (binding)
Passing grade	5.5

20.3. Resits and late submission

Failing to submit at the first deadline is treated the same as missing an exam — you have used one exam opportunity.

Situation	What happens
Not submitted at first deadline	First exam opportunity used; one resit available
Submitted but grade < 5.5	One resit available
Resit submitted but grade < 5.5	Failed the course; may be eligible for a second resit in the next academic year (see below)
Not submitted at first deadline AND resit	Failed the course; may be eligible for a second resit in the next academic year

20.3.1. Second resit (next academic year)

In limited circumstances, you may be eligible for a second resit in the following academic year (deadline: end of November). This requires re-enrollment and tuition fee payment. Eligibility is assessed by me and the thesis coordinator. Contact your study advisor.

20.4. Where to find formal regulations

The formal *Regulation Ma thesis* governs all MA theses at the Faculty of Social Sciences. It is available on VUnet. For the vast majority of students, these regulations remain in the background — we find a good working relationship without needing to consult them. They become relevant if serious disagreements arise.

20.5. Getting help

Need	Contact
Academic/personal difficulties	Study advisor: vu.nl/en/student/contact-student-guidance-and-support/academic-advisors-fss
Financial support for delays	Profileringsfonds: vu.nl/en/student/finances/profile-fund
Ethics self-check questions	rerc.ssc@vu.nl

21. Graduation

21.1. Graduation procedure

Once your thesis has a passing grade and you have completed all other courses:

1. You will automatically receive a graduation email within approximately two weeks
2. **Upload your thesis to the UBVU thesis database** — this is mandatory for receiving your diploma
    - Choose *Publicly available* under access rights
    - Instructions: vu.nl/en/about-vu/divisions/university-library/more-about/uploading-your-thesis
3. Indicate your attendance at the **graduation ceremony** (usually in October/November)

> **i** No defense required
>
> There is no oral presentation or defense of the MA thesis in Communication Science. The diploma ceremony is a celebration, not an assessment event.

21.2. Getting help

Need	Contact
Graduation questions	vu.nl/en/student/graduation-and-diploma
Academic/personal difficulties	Study advisor: vu.nl/en/student/contact-student-guidance-and-support/academic-advisors-fss
Financial support for delays	Profileringsfonds: vu.nl/en/student/finances/profile-fund